# TOWARDS NIELSEN–THURSTON CLASSIFICATION FOR SURFACES OF INFINITE TYPE

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Abstract. We introduce and study tempered homeomorphisms of surfaces of infinite type. These are maps for which curves under iterations do not accumulate onto geodesic laminations with non-proper leaves, but rather just a union of possibly intersecting curves or proper lines. Assuming an additional finiteness condition on the accumulation set, we prove a Nielsen–Thurston type classification theorem. We prove that for such maps there is a canonical decomposition of the surface into invariant subsurfaces on which the first return is either periodic or a translation.

## 1. Introduction

In this paper, we study mapping classes–i.e., isotopy classes of self-homeomorphisms–of a connected, orientable surface $S$ of *infinite* type. For finite type surfaces, the Nielsen–Thurston classification theorem asserts that every mapping class is either periodic, reducible, or pseudo-Anosov. By Nielsen realization, a periodic class is represented by an isometry of some hyperbolic metric on $S$. On the other hand, a pseudo-Anosov map preserve a pair of transverse geodesic laminations that are minimal and filling. In the reducible case, $S$ decomposes along an invariant multicurve so that the first return map to each complementary subsurface is either periodic or pseudo-Anosov.

The long-term goal of this project is to extend this understanding to surfaces of infinite type. In this paper, we take a step in that direction: we introduce the notion of *tempered* maps and prove a structure theorem for the subclass of *well tempered* maps.

To motivate our definitions, we recall several approaches to the Nielsen–Thurston theorem and assess their prospects in the infinite setting. Thurston's original proof [24, 1] is to find a fixed point in the compactified Teichmüller space. Bers' proof [9] also uses Teichmüller space, but from the point of view of extremal quasiconformal maps. Casson's proof [13] produces an invariant geodesic lamination by iterating a curve, while Nielsen's approach [19, 20, 21], completed by Miller [18] and Handel–Thurston [15], analyzes the dynamics on the circle at infinity of lifts of the map to the universal cover. The Bestvina–Handel proof [10] derives invariant train tracks from efficient spines of the surfaces and relies on Perron–Frobenius theory.

In the infinite-type setting, Teichmüller space is far more complicated: it is infinite-dimensional, has uncountably many connected components, and some maps do not preserve any component. So it is hard to imagine how Thurston's or Bers' approach could be adapted. Likewise, train-track methods would require infinite-sized matrices without a suitable analog of Perron–Frobenius theory. By contrast, the viewpoints of Casson and Nielsen–Handel–Thurston seem more amenable; in this paper, we take Casson's approach.

Given a map $f$, the first step in Casson's program is to construct a geodesic lamination $\lambda$ as the limit of a subsequence $f^{n_i}(\alpha)$ of iterates of a curve $\alpha$, pulled tight. To ensure that

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$f^n(\lambda)$ has no transverse intersections with $\lambda$ (in which case $\overline{\bigcup_n f^n(\lambda)}$ is an $f$-invariant geodesic lamination), this limiting process should be "robust", meaning that more and more strands of $f^{n_i}(\alpha)$ converge to the leaves of $\lambda$. This robustness is essentially automatic in the finite-type setting, and it fails only when $\alpha$ intersects a subsurface on which $f$ is periodic.

For infinite-type surfaces, several new phenomena appear, and the robustness discussed above is far from being automatic. For example, there are infinite-order isometries, such as the translation $(x, y) \mapsto (x+1, y)$ on the surface $\mathbb{R}^2 \smallsetminus \mathbb{Z}^2$, which sends every curve to infinity. This is an instance of a general *translation*–a map that generates an infinite cyclic group acting properly on the surface. For a more complicated example, consider the same surface but let the map agree with the translation on $y \leq 1/3$, its inverse on $y \geq 2/3$, and a continuous interpolation of the two maps by horizontal translations in the strip $1/3 < y < 2/3$. In this case, a curve that intersects the strip essentially will converge to a line homotopic into the strip, but the convergence is not robust. A different phenomenon is an *irrational rotation*. Start with a homeomorphism of the equator in $S^2$ whose minimal invariant set is a Cantor set $C$, and extend it to a homeomorphism $f$ of $S^2 \smallsetminus C$. Here, a convergent subsequence $f^{n_i}(\alpha)$ limits onto a line in a non-robust way, and the union of all such limits is a non locally-finite collection of lines, some of which intersect; in particular, it's not a geodesic lamination.

This brings us to the notion of *tempered* maps, which are precisely those maps where robust convergence never occurs for any curve; in other words, a tempered map does not exhibit pseudo-Anosov behavior anywhere on the surface. Concretely, a map $f : S \to S$ is tempered if for any two curves $\alpha, \beta$ on $S$, there is a uniform bound on the intersection numbers $i(f^n(\alpha), \beta)$ for all $n \in \mathbb{Z}$ (see Section 4). As shown in Section 4, under a tempered map $f$, a curve $\alpha$ either leaves every compact set, or has limit set $\mathcal{L}(\alpha)$ consisting of curves or properly embedded lines (rather than more complicated laminations). A tempered map is *well tempered* if, in addition, the limit set of every curve is a *finite* collection of lines and curves.

On a finite-type surface, being tempered is equivalent to periodicity. For infinite-type surfaces, prototypes of tempered maps are periodic maps, translations, and irrational rotations, with the first two being well tempered. More complicated examples can be built by gluing multiple maps together in a controlled way, such as the gluing of two translations along a fixed line like our second example above, and the same procedure can be extended to infinitely many maps.

Despite the name, tempered maps can exhibit complicated behavior. Nevertheless, we believe that there is a Nielsen–Thurston type theory for tempered maps and conjecture that our fundamental examples are the only building blocks of such maps. Our main result is a proof of this conjectural picture for well tempered maps. As we shall explain below, the full strength of our theorem cannot be generalized to all tempered maps.

**Theorem A.** *Let $f$ be a well tempered map of a surface $S$ of infinite type. Then, up to modifying $f$ by an isotopy, there is a hyperbolic metric on $S$ and a canonical decomposition of $S$ into three $f$–invariant subsurfaces $S_{per}$, $S_\infty$ and $S_0$ with totally geodesic boundary such that, for every component $X$ of $S_{per}$, $S_\infty$ and $S_0$, the map $f$ returns to $X$ and the first return is isotopic to:*

- *a periodic isometry if $X \subseteq S_{per}$ or $X \subseteq S_0$;*
- *an isometric translation if $X \subseteq S_\infty$.*

*Furthermore, each component $X$ of $S_0$ is topologically trivial, i.e. $X$ contains no essential non-peripheral curves and at most one peripheral curve.*

A component of $S_0$ should be regarded as the transition piece interpolating between the dynamics of on neighboring pieces. For instance, if $f$ is obtained from three translations with disjoint supports as in Figure 22 (instead of two as in the example described above), then $S_0$ has a component homeomorphic to an ideal triangle.

For general tempered maps, the statement of Theorem A must first be adjusted to accommodate irrational-rotation behavior. Even then, a full analogue of the theorem cannot hold. The main obstruction is that, for some tempered maps, the natural invariant pieces are no longer subsurfaces; although these pieces have non-empty interior, their boundaries accumulate in a complicated way (see the *Strategy of proof* section for more details). Even replacing finiteness of limit sets by local finiteness is not enough to resolve this issue (see Section 7.1).

When we first wrote this article, we proposed the following conjecture for general tempered maps.

**Conjecture B.** *Let $f$ be a tempered map of a surface $S$ of infinite type. Then, up to modifying $f$ by an isotopy, there is a canonical $f$-invariant lamination $\lambda$ such that for every component $X$ of $S \setminus \lambda$, if a first return map exists, it is*

- *periodic or a translation if $X$ is topologically nontrivial;*
- *periodic or an irrational rotation if $X$ is topologically trivial.*

*Moreover, all leaves of $\lambda$ are proper.*

In a forthcoming paper, we prove a slightly weakened version of Conjecture B: we show the existence of an invariant lamination such that, for every complementary component, if a first return map exists, it is either periodic, an *almost* translation or an irrational rotation. Here an almost translation is a map so that, for every two curves $\alpha$ and $\beta$, the set $\{n \mid i(f^n(\alpha), \beta) \neq 0\}$ has zero density in $\mathbb{Z}$. An irrational rotation is Denjoy map on either a closed disk with a Cantor set removed from the boundary, a once-punctured such disk or an annulus with a Cantor set removed from each boundary component. The question of whether the leaves of the lamination are proper remains open.

If $f$ is not tempered, then one could try to construct an $f$–invariant geodesic lamination using the Casson technique. This approach has been successfully carried out for the class of *irreducible end-periodic* maps by Handel and Miller [12]. In our vision, a comprehensive structure theorem for an arbitrary map $f$ should begin by decomposing $S$ into $f$–invariant pieces on which the first return is either tempered or preserves two transverse geodesic laminations. Therefore, a structure theorem for tempered maps is an essential part of the broader program, and Theorem A is a first step.

**Strategy of proof.** Given a map $f$ on $S$, our general strategy for decomposing $S$ into $f$–invariant pieces is to organize curves by their dynamical behavior under $f$, and then analyze the subsurface *spanned* by curves of a given type, when it exists. For instance, let $\mathcal{C}_{\mathrm{per}}$ be the collection of $f$–periodic curves. If the subsurface $S_{\mathrm{per}}$ spanned by $\mathcal{C}_{\mathrm{per}}$ exists, then $S_{\mathrm{per}}$ is $f$–invariant and the restriction of $f$ to each component of $S_{\mathrm{per}}$ is periodic (see Section 9).

We then apply the same idea to the collection $\mathcal{C}_\infty$ of *wandering* curves, i.e. curves that leave every compact set of $S$ under iteration. With some work, one can show that if the subsurface $S_\infty$ spanned by $\mathcal{C}_\infty$ exists, then the action of $f$ on each component of $S_\infty$ is by a translation. This characterization can be viewed as an analogue of Brouwer's plane translation theorem ([11]) and is proved in Section 8.

**Theorem C.** *Suppose $f$ is a homeomorphism of a surface $S$ such that every curve of $S$ is wandering. Then there is a hyperbolic metric on $S$ for which $f$ is isotopic to an isometric translation.*

When $f$ is tempered and the components $S_{\rm per}$ and $S_\infty$ exist, then the remaining step in proving Conjecture B is to show that $S \smallsetminus (S_{\rm per} \cup S_\infty)$ admits a further decomposition into $f$–invariant pieces of trivial topology, obtained by cutting along an $f$–invariant lamination with proper leaves. For a topologically trivial component, it then follows immediately that the first-return map–when it exists–is either periodic or has irrational-rotation behavior. In the well tempered case, irrational rotations do not occur, so it suffices to show that the components of $S \smallsetminus (S_{\rm per} \cup S_\infty)$ are topologically trivial and that $f$ always returns to each component.

As this strategy suggests, a key technical step is establishing the existence of the subsurface spanned by a collection of curves. For a finite collection of curves, the span is always well-defined: put the curves in general position, take the union of their regular neighborhoods, and cap off any disk components. For an infinite collection, one may take the span $S_n$ of the first $n$ curves and study the limiting object as $n \to \infty$. When the underlying surface $S$ has finite type, Euler characteristic considerations force the topology of $S_n$ to stabilize, and this procedure recovers the Nielsen–Thurston decomposition of $f$. For infinite-type surfaces, however, the limiting object need not be a subsurface of $S$. Indeed, there are examples of tempered maps for which the collection of periodic or wandering curves fail to span a subsurface (see Section 7.1). This issue turns out to be a significant obstacle to proving the conjecture in full generality. However, under the assumption of being well tempered, this bad behavior disappears, allowing this fundamental step to be completed successfully.

**Proposition D.** *If $f$ is well tempered, then the collection of $f$–periodic curves spans a subsurface, and the collection of $f$–wandering curves spans a subsurface. Moreover, the complement of their union in $S$ is a subsurface containing no essential non-peripheral curves.*

Proposition D is proved in Section 7 and follows from Lemmas 7.2 and 7.4.

**Connection to actions on graphs.** A related, and perhaps more tractable, variant of the Nielsen-Thurston classification theorem is to study the action of mapping classes on Gromov-hyperbolic graphs naturally associated to a surface, and to classify mapping classes by their dynamic types as graph isometries. For example, in the finite-type setting, pseudo-Anosov mapping classes act loxodromically on the curve graph of $S$, while all other mapping classes act elliptically.

In the infinite-type setting, the curve graph always has bounded diameter, reducing the classification of isometries to a trivial problem. However, many infinite-type surfaces admit other interesting Gromov-hyperbolic graphs of infinite diameter. One example is the ray graph, introduced by Bavard [8] for the plane minus a Cantor set $\mathbb{R}^2 \smallsetminus C$, generalized to surfaces with one isolated end by Aramayona–Fossas–Parlier [5], and extended further by Bar-Natan–Verberne [7], who define the *grand-arc graph*. In [8], Bavard also constructed a map of $\mathbb{R}^2 \smallsetminus C$ that acts loxodromically on the ray graph, and Abbott–Miller–Patel [2] generalized this construction to a large class of surfaces. However, a general classification of isometries remains open, and even the following question is unknown.

*Question* 1.1. Are there maps of $\mathbb{R}^2 \smallsetminus C$ that act as parabolic isometries of the ray graph? More generally, are there maps that act as parabolic isometries of the grand arc graphs?

By our work above, well tempered map are elliptic isometries of the ray graph, and our conjectural picture of tempered maps predicts that they are also elliptic isometries.

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## 2. Background

**Standing assumptions.** By a surface $S$ we will mean a connected orientable 2-manifold, without boundary unless otherwise stated. When we need to consider surfaces with boundary, we will usually say *bordered* surface for emphasis. By a *hyperbolic metric* on $S$ we always mean a complete hyperbolic metric without funnels or half-planes; equivalently, the hyperbolic surface coincides with its convex core, which is also equivalent to the metric being of the *first kind* – see [4]). For bordered surfaces, hyperbolic metrics are assumed to contain no funnels or half planes and to have totally geodesic boundary.

A surface $S$ has *finite type* if $\pi_1(S)$ is finitely generated and *infinite type* otherwise. By the classification of surfaces, $S$ is of finite type if and only if $S$ is homeomorphic to a closed genus $g$ surface with finitely many points removed. A *pair of pants* is a (possibly bordered) surface homeomorphic to either a three-holed sphere, or a two-holed once-punctured sphere, or a twice-punctured disk.

Associated to $S$ is its *ends*, which can be *planar* or *nonplanar*. A *puncture* is an isolated planar end. We refer to [6] for definitions and properties of ends of a surface.

The *mapping class group* of $S$ is the group $\mathrm{Map}(S)$ of isotopy classes of orientation-preserving homeomorphisms of $S$.

*Curves* on $S$ are assumed to be simple, closed, and not homotopic to a point or a puncture. We will sometimes call such a curve *essential* for emphasis. If $S$ has boundary, then a curve is called *peripheral* if it is homotopic to a boundary component. When $S$ is endowed with a hyperbolic metric with geodesic boundary, every essential curve is realized by a unique geodesic $\alpha^*$ representing its homotopy class.

A *line* on $S$ is the image of a proper embedding of $\mathbb{R} \to S$. We will usually assume a line is *essential*, meaning it does not bound a (topological) half-plane. We will usually consider lines up to proper isotopy relative to their ends. Similarly to curves, essential lines admit geodesic representatives:

**Lemma 2.1.** *Let $\ell$ be an essential line on a surface $S$ endowed with a hyperbolic metric (of the first kind). Then $\ell$ is isotopic to a unique geodesic $\ell^*$.*

*Sketch of proof.* Fix a pants decomposition of $S$, which yields an exhaustion of $S$ by finite-type subsurfaces with compact totally geodesic boundary (see [4]). Pick a lift $\tilde{\ell}$ of $\ell$ in the universal cover. Since $\ell$ is proper and the metric is of the first kind, by looking at the lifts of the pants curves we can conclude that $\tilde{\ell}$ has two distinct limit points on the circle at infinity. Let $\tilde{\ell}^*$ be the unique geodesic joining the two limit points and let $\ell^*$ be the projection of $\tilde{\ell}^*$ to $S$. Arguing as in the proof of Proposition 3.3, one shows that $\ell$ is isotopic $\ell^*$. Uniqueness follows because the limit points of $\tilde{\ell}$ determine $\tilde{\ell}^*$ and hence $\ell^*$. □

An *arc* on $S$ is the image of an embedding $[0,1] \to S$; in particular in our definition arcs are compact. If $S$ has boundary, an arc with endpoints on $\partial S$ is called *essential* if it doubles to an essential curve in the double of $S$.

A *ray* is the image of $[0,\infty)$ under a proper embedding.

A *geodesic lamination* is a closed subset of $S$ that is a union of complete, simple, and pairwise disjoint geodesics (with respect to some hyperbolic metric on $S$), called the *leaves* of the lamination. A simple closed geodesic or a geodesic line is a geodesic lamination with a single leaf.

The *geometric intersection number* $i(\alpha, \beta)$ is the minimum number of intersections between representatives of $\alpha$ and $\beta$ in their homotopy classes, where each of $\alpha$ and $\beta$ could be either a line, curve, or a leaf of a lamination. Note that $i(\alpha, \beta)$ could be infinite, but we have the following characterization of lines.

**Lemma 2.2.** *Let $S$ be a surface equipped with a hyperbolic metric. Then a complete simple geodesic $\ell \subset S$ is a line or a curve if and only if $i(\ell, \alpha) < \infty$ for all curves $\alpha$.*

*Proof.* Clearly, if $i(\ell, \alpha) = \infty$ for some curve $\alpha$, then $\ell$ cannot be proper. Conversely, suppose $i(\ell, \alpha) < \infty$ for all curves $\alpha$. Let $K \subset S$ be a compact subsurface. Since $\sum_{\alpha \subset \partial K} i(\ell, \alpha) < \infty$, $\ell \cap K$ has a finite number of components. If a component $\tau \subset \ell \cap K$ had infinite length, then $\tau$ would accumulate onto some geodesic lamination $\lambda \subset K$, or a boundary component $\gamma$ of $K$. In either case, any curve $\alpha$ intersecting $\lambda$ (or $\gamma$) would satisfy $i(\ell, \alpha) \geq i(\tau, \alpha) = \infty$. Thus $\ell \cap K$ is always a finite union of arcs, and hence $\ell$ is proper. □

2.1. **Subsurfaces.** A *subsurface* of $S$ is a closed subset $X \subset S$ that is a two-dimensional (possibly disconnected) submanifold such that each boundary component of $X$ is either an essential curve or an essential line of $S$. We also require that no connected component of $X$ can be homotoped into another. In particular, no component of a subsurface can be a closed disk with at most one puncture. Moreover, $X$ is a subsurface if and only if $Y = \overline{S \setminus X}$ is also a subsurface. We will usually consider a subsurface up to isotopy.

A connected subsurface $X$ is *essential* if its double either has negative Euler characteristic or is of infinite type. Topologically, this rules out annuli and *strips* (i.e. closed disks with two points removed from the boundary).

Two subsurfaces in $S$ are *disjoint* if they have disjoint representatives. Similarly, a curve or line or geodesic lamination is disjoint from a subsurface if they have disjoint representatives. Note that in this convention, the boundary of a subsurface is considered disjoint from the subsurface.

Fix a hyperbolic metric on $S$. Because we will usually consider a subsurface up to isotopy, it is convenient to pick out a canonical representative within each isotopy class. To this end, we will introduce the notion of an *(almost) geodesic representative* of a subsurface $X \subset S$, subject to the hyperbolic metric on $S$, which serve as an *almost* canonical representative of the isotopy class of $X$.

We begin with representatives of the interiors of subsurfaces. First suppose that $X$ is connected. A *geodesic representative of the interior of* $X$ is an open set $X^\circ$ defined as follows:

- If $X$ is essential, then $X^\circ$ is the open set properly homotopic to $\mathrm{int}(X)$ whose metric completion is a hyperbolic surface with totally geodesic boundary, homeomorphic to $X$.
- If $X$ is an annulus or a strip with core curve/line $\alpha$, then $X^\circ$ is the interior of any closed regular neighborhood of $\alpha^*$, whose boundary components are proper curves or lines .

For a (not necessarily connected) subsurface $X$, a *geodesic representative of its interior* (also denoted $X^\circ$) is the union, over the connected components of $X$, of geodesic representatives of their interiors. By construction, $X^\circ$ is an open subset properly homotopic to $\mathrm{int}(X)$.

wondering if we ld define everything out annuli first and add annuli later. t now it's a little using. –jt

To get a representative for $X$, we proceed as follows. For each component $Y$ of $X$, let $\overline{Y^\circ}$ be the closure of $Y^\circ$, and set $\partial Y^\circ := \overline{Y^\circ} \smallsetminus Y^\circ$. When $Y$ is essential, then $\partial Y^\circ$ is a disjoint union of simple closed geodesics and geodesic lines; otherwise, $\partial Y^\circ$ consists of a pair of almost geodesic curves or lines, each homotopic to the core curve/line of $Y$. Set $\partial X^\circ := \bigcup_{Y \subset X} \partial Y^\circ$, and let $\partial_{sa} X^\circ$ consists of those components $\alpha$ of $\partial X^\circ$ such that both boundary components of a regular neighborhood of $\alpha$ are contained in $X^\circ$. We define an *almost geodesic representative* of $X$ by

$$X^* = \overline{X^\circ} \smallsetminus \bigcup_{\alpha \in \partial_{sa} X^\circ} N(\alpha)$$

where $N(\alpha)$ is an open regular neighborhood $\alpha$.

An almost geodesic representative is well-defined up to our choices of regular neighborhoods of the components in $\partial_{sa}(X^\circ)$ and the representatives of the inessential components of $X$ (and the hyperbolic metric on $S$). In particular, if $\partial_{sa}(X^\circ) = \emptyset$ and $X$ has no inessential components, then $X^*$ is a totally geodesic subsurface representing $X$. Moreover, $\overline{X^\circ}$ is homeomorphic to $X^*$ if and only if $\partial_{sa} X^\circ = \emptyset$.

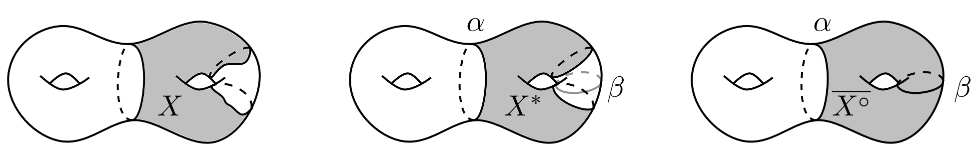


FIGURE 1. An example of a subsurface $X$ so that $\overline{X^\circ}$ is not homeomorphic to $X^*$. The curves $\alpha$ and $\beta$ are the geodesic representatives of the boundary components of $X$.

**Lemma 2.3.** *For any hyperbolic metric on $S$, any subsurface is properly homotopic to its almost geodesic representative.*

*Sketch of proof.* If $\partial_{sa} X^\circ = \emptyset$ and $X$ has no inessential components, then the almost geodesic representative agrees with the geodesic representative of $X$, and the claim follows by properly homotoping all boundary components of $X$ at once to their geodesic representatives (see Lemma 2.1 and [13, Lemma 2.6]. If $\partial_{sa} X^\circ \neq \emptyset$ or $X$ has annular or strip components, then the same argument applies with a minor modification to handle pairs of parallel boundary components. □

A consequence is that "inclusion up to proper homotopy" defines a partial order on the set of equivalence classes of subsurfaces defined up to proper homotopy:

**Corollary 2.4.** *Let $S$ be a surface equipped with a hyperbolic metric. If $X_1$ and $X_2$ are subsurfaces such that each can be properly homotopic into the other, then $X_1$ and $X_2$ are properly homotopic.*

*Proof.* We first prove that inessential components of $X_1$ and $X_2$ are the same, up to proper homotopy. Indeed, suppose $Y$ is a inessential component of $X_1$. Since $X_1$ is properly homotopic into $X_2$, $Y$ properly homotopes into some component $Z$ of $X_2$. Since $X_2$ properly homotopes into $X_1$, $Z$ is properly homotopic into some component $Y'$ of $X_1$. Thus $Y$ properly homotopes into $Y'$, and by the definition of subsurface (no component homotopes into another), we must have $Y = Y'$. In particular, $Z$ properly homotopes into a strip or an annulus and is hence a strip or an annulus itself. By repeating the same argument for the inessential components of $X_2$ we get the claim.

We may therefore assume that $X_1$ and $X_2$ have no inessential components. With respect to the hyperbolic metric on $S$, $X_1^\circ$ and $X_2^\circ$ are uniquely defined. We claim they are the same. Otherwise, after swapping indices if necessary, there exists a component $Y$ of $X_1^\circ \smallsetminus X_2^\circ$. Since $X_1$ properly homotopes into $X_2$, such $Y$ can only be a curve or a line, so $Y \subset \partial_{sa} X_1^\circ$. Choose an essential curve or line $\alpha \subset X_1^\circ$ that intersects $Y$ essentially. Again, because $X_1$ properly homotopes into $X_2$, $\alpha$ can be homotoped into $X_2$ and hence must be disjoint from $Y$, a contradiction. This shows $X_1^\circ = X_2^\circ$, so $X_1$ and $X_2$ have the same almost geodesic representative. Lemma 2.3 then implies that $X_1$ and $X_2$ are properly homotopic. □

2.2. **Some results on isometries of surfaces.** We say that a collection $\mathcal{C}$ of curves *cuts* a (sub)surface $X$ if every non-peripheral curve in $X$ has positive intersection number with some curve in $\mathcal{C}$. The following well-known statement follows from the proof of [13, Theorem 2.7].

**Lemma 2.5.** *Let $F$ be a possibly bordered finite-type surface of negative Euler characteristic, whose boundary is either empty or compact. Let $f$ be a mapping class of $F$. If $F$ is cut by a finite collection of $f$–periodic curves, then $f$ has finite order.*

*Remark* 2.6. The order of $f$ in Lemma 2.5 can be arbitrarily larger than the periods of the cutting curves. For instance, consider the torus $\mathbb{R}^2/\mathbb{Z}^2$ and the curves $\alpha$ and $\beta$ given by the projection of the lines $y = nx$ and $y = -nx$, for some positive integer $n$. Puncture the torus at the points that are the projections of $\left(\frac{1+2k}{2n}, 0\right)$ and $\left(\frac{k}{n}, \frac{1}{2}\right)$, for $k = 0, \ldots, n-1$, which do not meet $\alpha$ and $\beta$. Then the map induced by $(x, y) \mapsto \left(x + \frac{1}{2n}, y + \frac{1}{2}\right)$ has order $2n$, but it fixes both $\alpha$ and $\beta$.

We will also need a realization result for periodic mapping classes of infinite-type surfaces; the proof is essentially borrowed from Afton-Calegari-Chen-Lyman ([3]).

**Proposition 2.7.** *Let $S$ be an infinite-type surface with boundary, and $f$ a homeomorphism of $S$ such that $f^n$ homotopic to the identity for some $n > 0$. For each compact boundary $\alpha$ of $S$, choose a positive number $\ell(\alpha)$ that is $f$–invariant, i.e. $\ell(f(\alpha)) = \ell(\alpha)$. Then there exists a hyperbolic metric on $S$ such that each compact boundary component $\alpha$ has length $\ell(\alpha)$ and $f$ is isotopic to a periodic isometry of $S$.*

*Proof.* Suppose first that $S$ has only compact boundary components. Using Nielsen's realization theorem for finite-type surfaces with compact boundary [22] and the argument of Afton-Calegari-Chen-Lyman (see the proof of [3, Theorem 2]), one obtains the desired metric and periodic representative in this case.

Now suppose $S$ has noncompact boundary components. Let $D(S)$ be the double of $S$ along its noncompact boundary and extend $f$ to a homeomorphism $\hat{f}$ of $D(S)$. Let $i$ be the involution of $D(S)$ with quotient $S$, which commutes with $\hat{f}$ by construction. By [3] and the discussion above, there is an $i$–invariant metric on $D(S)$ realizing the prescribed lengths for the compact components, and a periodic isometry $g$ isotopic to $\hat{f}$ that commutes with $i$. Since $i$ is an isometry, the noncompact boundaries of $S$ are realized as geodesics in $D(S)$, so that the metric descends to a metric on $S$ with totally geodesic boundary. Since $g$ commutes with $i$, it descends to a periodic isometry of $S$ isotopic to $f$. □

## 3. Limit sets

Throughout this section, $S$ is a surface without boundary equipped with a hyperbolic metric, though everything we say will be independent of the particular choice of metric. Indeed, the following holds (see [25, Theorem 3.6] for the infinite-type case).

**Theorem 3.1.** *Let $m$ and $m'$ be two hyperbolic metrics on $S$. The identity map on $S$ induces a natural identification between the boundaries at infinity of the universal covers of $(S, m)$ and $(S, m')$. This identification yields a homeomorphism between the spaces of complete geodesics in the universal covers of $(S, m)$ and $(S, m')$, which descends to a homeomorphism between complete $m$-geodesics and complete $m'$-geodesics on $S$.*

Given a complete $m$-geodesic $\ell$, we call the corresponding $m'$-geodesic the *$m'$-straightening* of $\ell$. For a collection of $m$-geodesics, its *$m'$-straightening* is the union of the $m'$-straightenings of its geodesics. A consequence of Theorem 3.1 is the following:

**Lemma 3.2.** *Let $m$ and $m'$ be two hyperbolic metrics on $S$. If two complete $m$–geodesics $\ell_1$ and $\ell_2$ are simple and disjoint, then the $m'$–straightenings $\ell_1$ and $\ell_2$ are also simple and disjoint. If $\lambda$ is an $m$–geodesic lamination, its $m'$–straightening is an $m'$–geodesic lamination.*

The reason why the lemma holds is that (self-)intersections correspond to linked pairs of endpoints at infinity of lifts and convergence of geodesics corresponds to convergence of pairs of endpoints at infinity of lifts. Moreover, we have:

**Proposition 3.3.** *Let $m$ and $m'$ be hyperbolic metrics on $S$ and $\lambda$ an $m$-geodesic lamination. Then there is a homeomorphism of $S$, isotopic to the identity, that maps $\lambda$ leaf-by-leaf to its $m'$-straightening $\lambda'$.*

*Proof.* Pick an $m$–geodesic pants decomposition $\mathcal{P}$ that contains no leaf of $\lambda$ (this can be done by choosing a pants decomposition and performing elementary moves until none of its curves lie in $\lambda$). In particular, $\mathcal{P}$ is in minimal position with respect to $\lambda$. Let $\mathcal{P}'$ be the $m'$–straightening of $\mathcal{P}$. There is an ambient isotopy $H_1$ of $S$ sending $\mathcal{P}$ to $\mathcal{P}'$. Let $\lambda_1 := H_1(\lambda)$. Then $\mathcal{P}'$ is in minimal position with respect to both $\lambda_1$ and $\lambda'$, and contains no leaf of either.

Now pick a curve $\alpha$ in $\mathcal{P}'$, and lift $\alpha$, $\lambda_1$, and $\lambda'$ to the universal cover of $(S, m')$. Choose and orient a lift $\beta$ of $\alpha$. Given a lift $\ell_1$ of a leaf in $\lambda_1$, its $m'$–straightening is the unique lift $\ell_1'$ of the $m'$–straightened leaf in $\lambda'$ with the same endpoints; in particular, $\ell_1$ intersects $\beta$ if and only if $\ell_1'$ intersects $\beta$. Moreover, if $\ell_1$ and $\ell_2$ are two such lifts intersecting $\beta$, and $\ell_1'$ and $\ell_2'$ are their $m'$–straightenings, then the intersection points $\ell_1 \cap \beta$ and $\ell_2 \cap \beta$ occur in the same order along $\beta$ as $\ell_1' \cap \beta$ and $\ell_2' \cap \beta$. Hence there is an order-preserving homeomorphism

$$\tilde{\lambda_1} \cap \beta \to \tilde{\lambda}' \cap \beta$$

sending $\ell \cap \beta$ to $\ell' \cap \beta$, for every leaf $\ell \subset \tilde{\lambda_1}$, where $\ell'$ is its $m'$-straightening. Extending this to an equivariant isotopy on a small neighborhood of $\beta$, and doing so equivariantly for every lift of every curve $\alpha$ in $\mathcal{P}'$ on pairwise disjoint neighborhoods, produces an equivariant isotopy of the universal cover. This descends to an ambient isotopy $H_2$ of $S$, sending $\lambda_1$ to a lamination $\lambda_2$, such that $\lambda_2 \cap \alpha = \lambda' \cap \alpha$ for all $\alpha$ in $\mathcal{P}'$.

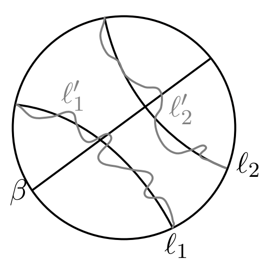


FIGURE 2. Lifting a curve in $\mathcal{P}'$

Now let $P$ be a pair of pants of $\mathcal{P}'$. Since $\lambda_2$ and $\lambda'$ are transverse to $\partial P$, $\lambda_2 \cap P$ and $\lambda' \cap P$ are unions of arcs with endpoints on $\partial P$. We claim that there is an isotopy of $P$, fixing $\partial P$ pointwise, sending $\lambda_2 \cap P$ to $\lambda' \cap P$. Partition the arcs of $\lambda_2 \cap P$ into (at most three) homotopy classes relative to the boundary. Do the same for the arcs of $\lambda' \cap P$; note that the same classes occur for both. For a class of $\lambda_2 \cap P$, choose a rectangle $R \subset P$, whose two opposite sides lie in $\partial P$ and whose two other sides are distinct arcs of $\lambda_2 \cap P$, such that $R$ contains all arcs of $\lambda_2 \cap P$ in that class. Construct a corresponding rectangle $R' \subset P$ for the same class of $\lambda' \cap P$, arranged so that $R$ and $R'$ have the same four corners on $\partial P$. Then an isotopy of $P$ fixing $\partial P$ pointwise can be chosen to send $R$ to $R'$ and the arcs of $\lambda_2 \cap R$ to the corresponding arcs of $\lambda' \cap R'$. Doing this for each homotopy class of arcs yields the desired isotopy of $P$.

Repeating the procedure in each pair of pants produces an ambient isotopy $H_3$ of $S$, fixing each curve of $\mathcal{P}'$ pointwise, and sending $\lambda_2$ to $\lambda'$. The composition $H_3 \circ H_2 \circ H_1$ is the required isotopy. $\square$

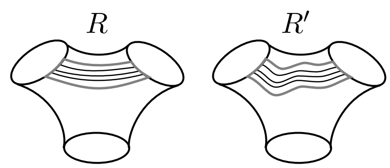


FIGURE 3. Isotopying rectangles

Let $\lambda \subset S$ be a closed subset. We say that a sequence of curves $\{\alpha_n \mid n \in \mathbb{N}\}$ *converges* to $\lambda$, and that $\lambda$ is the *limit* of $\{\alpha_n \mid n \in \mathbb{N}\}$, if for every finite-type subsurface $K \subset S$ with compact boundary, $\alpha_n^* \cap K$ converges to $\lambda \cap K$ with respect to the Hausdorff distance. We will also denote this by $\alpha_n \to \lambda$. The same definition applies to sequences of geodesic arcs. Since convergence is tested on finite-type subsurfaces, the standard argument (see e.g. [13]) implies:

**Lemma 3.4.** *Let $\lambda \subset S$ be a closed subset. If $\lambda$ is the limit of a sequence of curves $\{\alpha_n \mid n \in \mathbb{N}\}$, then $\lambda$ is a geodesic lamination. If $\lambda$ is the limit of a sequence of geodesic arcs, it is a union of pairwise disjoint geodesic arcs, geodesic rays and/or complete simple geodesics.*

The *limit set* of a sequence of curves $\{\alpha_n \mid n \in \mathbb{N}\}$ is the set $\mathcal{L}(\{\alpha_n \mid n \in \mathbb{N}\})$ consisting of all complete geodesics that occur as leaves of the limit of some subsequence of $\{\alpha_n \mid n \in \mathbb{N}\}$. Similarly we can define the limit set of a sequence of geodesic arcs. Changing the metric on $S$ does not change the limit set, in the following sense.

**Lemma 3.5.** *Let $m$ and $m'$ be two hyperbolic metrics on $S$. For a sequence $\{\alpha_n\}$ of curves, let $\mathcal{L}(\{\alpha_n \mid n \in \mathbb{N}\}, m)$ and $\mathcal{L}(\{\alpha_n \mid n \in \mathbb{N}\}, m')$ be their respective limit sets in the two metrics. Then a subsequence $\{\alpha_{n_j}\}_{j\in\mathbb{N}}$ converges to $\lambda \subset \mathcal{L}(\{\alpha_n \mid n \in \mathbb{N}\}, m)$ if and only if $\{\alpha_{n_j}\}_{j\in\mathbb{N}}$ converges to $\lambda' \subset \mathcal{L}(\{\alpha_n \mid n \in \mathbb{N}\}, m')$, where $\lambda'$ is the $m'$–straightening of $\lambda$.*

*Proof.* As convergence of geodesics corresponds to convergence of pairs of endpoints at the boundary at infinity, this follows from Theorem 3.1. $\square$

This justifies our notation $\mathcal{L}(\{\alpha_n \mid n \in \mathbb{N}\})$, which makes no reference to the metric on $S$.

3.1. **Limit sets under a homeomorphism.** For a homeomorphism $f$ of $S$ and a curve $\alpha$, we say $\alpha$ is *$f$-periodic* if there exists $n \geq 1$ such that $f^n(\alpha)$ is isotopic to $\alpha$. The smallest such $n$ is called the *$f$-period* of $\alpha$. We say $\alpha$ is *$f$–forward wandering* if $\{f^n(\alpha)^*\}_{n\geq 0}$ leaves every compact set of $S$, and $f$-*backward wandering* if it is $f^{-1}$-forward wandering. The curve $\alpha$ is $f$-*wandering* if it is both $f$-forward and $f$-backward wandering. These properties are independent of the hyperbolic metric on $S$ as well as perturbing $f$ by an isotopy. Thus, we will also adopt the same definition for a mapping class of $S$. We will usually suppress the reference to $f$ when the context is clear.

Note that a curve is $f$-wandering if and only if for every compact set $C \subset S$, $f^n(\alpha)$ can be homotoped to be disjoint from $C$ for all sufficiently large $|n|$. With this in mind, we can extend the definition of $f$–wandering to lines or subsurfaces of $S$, by which we mean that for every compact set $C \subset S$, the images under $f^n$ can be homotoped away from $C$ for any $n$ with sufficiently large $|n|$.

For a curve $\alpha$, let $\mathcal{L}^+(\alpha) := \mathcal{L}(\{f^n(\alpha) \mid n \geq 0\})$ be the *forward limit* of $\alpha$ under $f$, and $\mathcal{L}^-(\alpha) := \mathcal{L}(\{f^n(\alpha) \mid n \leq 0\})$ the *backward limit*. Set $\mathcal{L}(\alpha) := \mathcal{L}^+(\alpha) \cup \mathcal{L}^-(\alpha)$. When $\alpha$ is $f$–periodic, then

$$\mathcal{L}^{\pm}(\alpha) = \{\alpha^*, f(\alpha)^*, \dots, f^{n-1}(\alpha)^*\},$$

where $n \geq 1$ is the $f$–period of $\alpha$. Moreover, $\alpha$ is forward wandering if and only if $\mathcal{L}^+(\alpha) = \emptyset$, and it is backward wandering if and only if $\mathcal{L}^-(\alpha) = \emptyset$. See Corollary 4.3.

We establish some basic facts.

**Lemma 3.6.** *Let $f$ be a homeomorphism and $\alpha$ a curve. Suppose $\lambda$ is the limit of a sequence of iterates $\{f^{n_j}(\alpha) \mid j \in \mathbb{N}\}$. Then for any curve $\beta$, $i(\beta, \lambda) \neq 0$ if and only if $i(\beta, f^{n_j}(\alpha)) \neq 0$ for all sufficiently large $j$. Moreover, if $i(\beta, \lambda) \geq k$, then $i(\beta, f^{n_j}(\alpha)) \geq k$ for all sufficiently large $j$.*

*Proof.* If we look at an annular neighborhood $K$ of $\beta$, $f^{n_j}(\alpha)^* \cap K$ converges to $\lambda \cap K$ in the Hausdorff metric, so the first statement follows. If $\beta$ intersects $\lambda$ at least $k$ times, then we can find $k$ arcs in $K \cap \lambda$. Choose sufficiently small neighborhood about each arc so that they are pairwise disjoint. Then for all sufficiently large $j$, $f^{n_j}(\alpha) \cap K$ will pass through each of these neighborhoods, so $i(\beta, f^{n_j}(\alpha)) \geq k$. ☐

**Lemma 3.7.** *Let $f$ be a homeomorphism. Then for any two (not necessarily distinct) curves $\alpha$ and $\beta$ we have:*

$$i(\alpha, \mathcal{L}^+(\beta)) \neq 0 \iff i(\mathcal{L}^-(\alpha), \beta) \neq 0.$$

*Proof.* Suppose $i\left(\alpha, \mathcal{L}^+(\beta)\right) \neq 0$. Let $n_j \to \infty$ be a sequence so that $f^{n_j}(\beta) \to \lambda \subset \mathcal{L}^+(\beta)$ with $i(\alpha, \lambda) \neq 0$. By Lemma 3.6, for every $j$ large enough,

$$i(\alpha, f^{n_j}(\beta)) \neq 0$$

and thus

$$i(f^{-n_j}(\alpha), \beta) \neq 0.$$

In particular, $\alpha$ is not backward wandering. By taking a further subsequence so that $f^{-n_{j_k}}(\alpha) \to \lambda' \subset \mathcal{L}^-(\alpha)$, we have, again by Lemma 3.6, $i(\lambda', \beta) \neq 0$, i.e. $i(\mathcal{L}^-(\alpha), \beta) \neq 0$. By replacing $f$ by $f^{-1}$ we get the other implication. ☐

**Lemma 3.8.** *Let $f$ be a homeomorphism. Then a curve $\alpha$ is forward wandering if and only if $i(\alpha, \mathcal{L}^-(\beta)) = 0$ for all curves $\beta$. Similarly, $\alpha$ is backward wandering if and only if $i(\alpha, \mathcal{L}^+(\beta)) = 0$ for all $\beta$.*

*Proof.* If $\alpha$ is not forward wandering, then we can choose a curve $\beta$ with $i(\mathcal{L}^+(\alpha), \beta) \neq 0$. By Lemma 3.7, this implies $i(\alpha, \mathcal{L}^-(\beta)) \neq 0$. Conversely, if $i(\alpha, \mathcal{L}^-(\beta)) \neq 0$, then again by Lemma 3.7, $i(\mathcal{L}^+(\alpha), \beta) \neq 0$. In particular $\mathcal{L}^+(\alpha) \neq \emptyset$, so $\alpha$ is not forward wandering. ☐

**Corollary 3.9.** *Let $f$ be a homeomorphism. If $\alpha$ is a periodic curve and $\beta$ is a wandering curve, $i(\alpha, \beta) = 0$.*

*Proof.* If $i(\alpha, \beta) \neq 0$, $i(\mathcal{L}^+(\alpha), \beta) \neq 0$ (as $\alpha \in \mathcal{L}^+(\alpha)$), so $\beta$ is not wandering by Lemma 3.8. □

## 4. Tempered maps

As usual, assume $S$ is a surface without boundary endowed with a hyperbolic metric. We say a homeomorphism $f$ of $S$ is *tempered* if for every finite-type subsurface $K \subset S$ with compact boundary and every curve $\alpha$ there are $N = N(K, \alpha) \in \mathbb{N}$ and finitely many isotopy (relative to $\partial K$) classes of arcs from $\partial K$ to $\partial K$ and curves in $K$ such that, for every $n \in \mathbb{N}$, the intersection $f^n(\alpha)^* \cap K$ has at most $N$ components and each is in one of the given isotopy classes. Being tempered can be characterized by looking at intersections of curves, as the following lemma shows.

**Lemma 4.1.** *Let $f$ be a homeomorphism of $S$. The following statements are equivalent.*

*(1) $f$ is tempered.*
*(2) $f^{-1}$ is tempered.*
*(3) For curves $\alpha$ and $\beta$, $i(f^n(\alpha), \beta)$ is uniformly bounded for all $n \geq 0$.*
*(4) For curves $\alpha$ and $\beta$, $i(f^n(\alpha), \beta)$ is uniformly bounded for all $n \leq 0$.*
*(5) For curves $\alpha$ and $\beta$, $i(f^n(\alpha), \beta)$ is uniformly bounded for all $n$.*

*In particular, being tempered is independent of the hyperbolic metric on $S$.*

*Proof.* The equivalence of (3), (4), and (5) are immediate as the action of $f$ preserves geometric intersection number. That (1) implies (3) follows from taking the compact surface to be an annular neighborhood of the curve $\beta$. On the other hand, if $f$ is not tempered, then there exists a finite-type subsurface $K$ with compact boundary, a curve $\alpha$, and a subsequence $\{n_j\} \subset \mathbb{N}$ such that $f^{n_j}(\alpha)^* \cap K$ has $N_j$ components with $N_j \to \infty$ and $j \to \infty$, or includes components with increasing multiplicity. Then there exists a curve $\beta \subset K$ such that $i(f^{n_j}(\alpha), \beta) \to \infty$. This shows the equivalence of (1) and (3), as well as (2) and (4) by symmetry, whence the equivalence of all statements. Property (5) is independent of the hyperbolic metric on $S$, so the same is true of being tempered. □

We say a mapping class $f \in \mathrm{Map}(S)$ is *tempered* if it has a tempered representative. From the characterization of Lemma 4.1, if $f$ has a tempered representative, then all of its representatives are tempered. We call a tempered homeomorphism $f$ *well tempered* if the limit set $\mathcal{L}(\alpha)$ is finite for every curve $\alpha$. This property is also inherited by the mapping class of $f$.

As a point of reference, on a finite-type surface the following are equivalent: being tempered, well tempered, periodic, and isotopic to an isometry for some hyperbolic structure. In the infinite-type case, the situation is more complicated. Being periodic implies being isotopic to an isometry (with respect to some hyperbolic structure), which implies being well tempered, and well tempered maps are tempered by definition. None of these implications is an equivalence.

For instance, consider a translation of a surface $S$, which — as mentioned in the introduction — is a map $f$ that generates an infinite cyclic group acting properly on $S$. If $S$ has infinite type, then the quotient surface $X = S/\langle f \rangle$ has negative Euler characteristic or has infinite type, so we can lift a hyperbolic metric from $X$ to $S$ so that $f$ acts by an (infinite-order) isometry. In this case, every curve on $S$ is wandering, so $f$ is (well) tempered.

Another example is what we call an *irrational rotation*. Start with a homeomorphism of the circle with minimal invariant subset a Cantor set $C$ (see the Denjoy construction [17, Section 12.2]). View the circle as the equator of the two-sphere $S^2$, and extend the map to a homeomorphism of $S = S^2 \setminus C$. This map is tempered but *not* well tempered: every curve has an infinite limit set, consisting of lines which may intersect. More complicated examples of (well) tempered maps can be found in Sections 7.1 and 9.

On the other hand, a Dehn twist is not tempered: if $\tau$ is the twist about a curve $\alpha$, let $\beta$ be a curve intersecting $\alpha$ essentially. If $K$ is a finite-type subsurface with compact boundary containing $\alpha$ and $\beta$, the curves $\tau^n(\beta) \cap K = \tau^n(\beta)$ are all distinct. Another example of a map which is not tempered is a homeomorphism which restricts to a pseudo-Anosov on some finite-type subsurface (with compact boundary).

The goal of the remainder of this section is to describe properties of limit sets of curves and lines under tempered homeomorphisms. Our first result is that if a curve or line is not periodic, then its limit set can only contain lines.

**Proposition 4.2.** *Let $f$ be a tempered homeomorphism and $\alpha$ a non-periodic curve. If $\alpha$ is not forward wandering, then $\mathcal{L}^+(\alpha)$ is a non-empty collection of lines. Similarly, if $\alpha$ is not backward wandering, then $\mathcal{L}^-(\alpha)$ is a non-empty collection of lines.*

*Proof.* Assume $\alpha$ is not forward wandering, and for convenience, write $\alpha_n = f^n(\alpha)^*$. Suppose $\mathcal{L}^+(\alpha) = \emptyset$, then for every compact subsurface $K \subset S$, the set $\bigcup_{n\geq 0} \alpha_n \cap K$ has no accumulation points. Equivalently, $\bigcup_{n\geq 0} \alpha_n \cap K$ consists of finitely many arcs or curves. But two geodesics that coincide along a non-trivial arc must be equal, and $\alpha$ is not $f$–periodic, we must have $\alpha_n \cap K = \emptyset$ for all sufficiently large $n$. But this contradicts the assumption that $\alpha$ is not forward wandering. By Lemma 3.4, $\mathcal{L}^+(\alpha)$ is a non-empty union of complete simple geodesics. If $\mathcal{L}^+(\alpha)$ contains a non-proper leaf $\ell$, then by Lemma 2.2 there exists a curve $\beta$ such that $i(\beta, \ell) = \infty$. Let $\lambda \subset \mathcal{L}^+(\alpha)$ be the limit of some subsequence $\{\alpha_{n_j}\}$ containing $\ell$. By Lemma 3.6 $i(f^{n_j}(\alpha), \beta) = i(\alpha_{n_j}, \beta) \to \infty$, contradicting the characterization of being tempered given by Lemma 4.1. If instead $\mathcal{L}^+(\alpha)$ contains a compact leaf $\beta$, then since $\mathcal{L}^+(\alpha)$ has no non-proper leaves, $\beta$ arises as the limit of some subsequence $\{\alpha_{n_i}\}$. By the tempered condition, such a sequence has to be eventually constant, but this contradicts the non-periodicity of $\alpha$. Therefore $\mathcal{L}^+(\alpha)$ has no compact or non-proper leaves, so it is a non-empty union of lines. By replacing $f$ by $f^{-1}$ we get the second statement. $\square$

As $\mathcal{L}^{\pm}(\alpha)$ is well defined for mapping classes, we obtain the following immediate corollary.

**Corollary 4.3.** *For a tempered mapping class $f$ and a curve or line $\alpha$, the following statements hold.*

*(1) $\mathcal{L}^+(\alpha) = \emptyset$ if and only if $\alpha$ is forward wandering.*
*(2) $\mathcal{L}^-(\alpha) = \emptyset$ if and only if $\alpha$ is backward wandering.*
*(3) $\mathcal{L}(\alpha) = \emptyset$ if and only if $\alpha$ is wandering.*
*(4) $\mathcal{L}(\alpha)$ contains a curve if and only if $\alpha$ is a periodic curve, in which case $\mathcal{L}(\alpha) = \mathcal{L}^+(\alpha) = \mathcal{L}^-(\alpha)$ is the orbit of $\alpha$ under $f$.*

Another consequence of being tempered is that limits of iterates of curves can intersect only finitely many times.

**Lemma 4.4.** *Let $f$ be a tempered homeomorphism. For any two (not necessarily distinct) curves $\alpha$ and $\beta$, if $f^{n_j}(\alpha) \to \lambda \subset \mathcal{L}(\beta)$ and $f^{m_k}(\beta) \to \lambda' \subset \mathcal{L}(\beta)$, for some sequences $\{n_j\}$ and $\{m_k\}$, then $i(\lambda, \lambda') < \infty$.*

*Proof.* If $i(\lambda, \lambda') = \infty$, then for every $N$ there are $j, k$ such that $i(f^{n_j}(\alpha), f^{m_k}(\beta)) \geq N$, by Lemma 3.6. Thus, $i(\alpha, f^{m_k - n_j}(\beta)) \geq N$, so $f$ is not tempered by Lemma 4.1. $\square$

## 5. Subsurfaces and spanning sets of curves

Throughout this section, we fix a hyperbolic structure on $S$. Given a collection of curves $\mathcal{C}$, we say a curve $\alpha$ is a *concatenation* of curves in $\mathcal{C}$ if it is homotopic to a finite concatenation of piecewise-geodesic arcs coming from the geodesic representatives of curves in $\mathcal{C}$. The collection $\mathcal{C}$ is *closed under concatenation* if every curve obtained as a concatenation of curves in $\mathcal{C}$ is again an element of $\mathcal{C}$.

We say that a subsurface $\Sigma$ is *spanned* by $\mathcal{C}$ if:

(1) every curve in $\mathcal{C}$ is contained in $\Sigma$, up to isotopy; and
(2) every curve $\gamma \subset \Sigma$ is a concatenation of the curves in $\mathcal{C}$

and $\Sigma$ is minimal (with respect to inclusion, up to proper homotopy) among all subsurfaces satisfying (1) and (2).

Note that if $\mathcal{C}$ spans a subsurface $\Sigma$, then necessarily $\mathcal{C}$ cuts $\Sigma$. On the other hand, a (sub)surface $\Sigma$ need not coincide with the subsurface spanned by its curves. For instance, on a once-punctured suface of genus two we can consider the subsurface $\Sigma$, homeomorphic to a one-holed torus with a point removed from the boundary, as in Figure 4. Here, the subsurface spanned by the curves of $\Sigma$ is a one-holed torus with compact boundary.

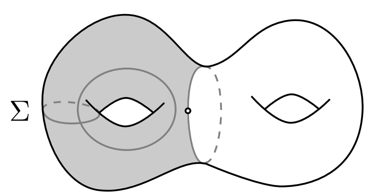


Figure 4. A subsurface $\Sigma$ not spanned by its curves.

We say that $X$ is *spanned by its curves* if it coincides with the subsurface spanned by its curves.

When $\mathcal{C}$ is a finite collection, there is always a subsurface $\Sigma$ (of finite type) spanned by $\mathcal{C}$. Namely, put the curves in $\mathcal{C}$ in general position, take a regular neighborhood of their union, and fill in any complementary disks with at most one puncture.

In general, one can show that if a subsurface spanned by a collection $\mathcal{C}$ of curves exists, then it is unique up to proper homotopy. To prove this fact, we introduce three subcollections of curves, which we will also use later:

$$\mathcal{C}_{\text{is}} := \{\gamma \in \mathcal{C} \mid \forall \gamma' \in \mathcal{C},\ i(\gamma, \gamma') = 0\};$$
$$\mathcal{C}_{\text{int}} := \mathcal{C} \setminus \mathcal{C}_{\text{is}};$$
$$\mathcal{C}_{\text{a}} := \{\gamma \in \mathcal{C}_{\text{is}} \mid \gamma \text{ is not a concatenation of curves in } \mathcal{C}_{\text{int}}\};$$

If a subsurface spanned by $\mathcal{C}$ exists, then the curves in $\mathcal{C}_{\text{a}}$ are precisely those that contribute annular components to the spanned subsurface, while the curves in $\mathcal{C}_{\text{is}} \setminus \mathcal{C}_{\text{a}}$ occur as boundary components of non-annular components.

**Lemma 5.1.** *Let $\mathcal{C}$ be a collection of curves in $S$. If $\Sigma_1$ and $\Sigma_2$ are subsurfaces spanned by $\mathcal{C}$, then they are properly homotopic.*

*Proof.* Fix a hyperbolic structure on $S$, and assume that all curves in $\mathcal{C}$ are realized as geodesics. Let $\Sigma_i^\circ$ be the geodesic representative of the interiors of $\Sigma_i$, chosen so that $\partial\Sigma_1^\circ$ and $\partial\Sigma_2^\circ$ are in minimal position. Let $\Sigma$ be an almost geodesic representative of $\Sigma_1^\circ \cap \Sigma_2^\circ$. Then $\Sigma$ is properly homotopic into both $\Sigma_1$ and $\Sigma_2$. We claim that $\Sigma$ contains all curves in $\mathcal{C}$ and that all curves in $\Sigma$ are concatenations of curves in $\mathcal{C}$. Then, by minimality of $\Sigma_1$ and $\Sigma_2$, they both are properly homotopic to $\Sigma$ and hence to each other.

It remains to prove the claim.

(1) Let $\gamma \in \mathcal{C}$.
- If $\gamma \notin \mathcal{C}_{\mathrm{is}}$, then $\gamma$ is a non-peripheral curve of a subsurface contained both in $\Sigma_i$. Hence it is contained in both $\Sigma_i^\circ$, and therefore $\gamma \subset \Sigma$, up to homotopy.
- If $\gamma \in \mathcal{C}_{\mathrm{a}}$, then $\gamma$ corresponds to an annular component contained in both $\Sigma_i$. Hence it is contained in both $\Sigma_i^\circ$, and therefore $\gamma \subset \Sigma$, up to homotopy.
- If $\gamma \in \mathcal{C}_{\mathrm{is}} \smallsetminus \mathcal{C}_{\mathrm{a}}$, then there are finitely many curves in $\mathcal{C} \smallsetminus \mathcal{C}_{\mathrm{is}}$, say $\beta_1, \dots, \beta_k$, such that $\gamma$ is a concatenation of arcs of the $\beta_i$. In particular, $\gamma$ is contained in the subsurface spanned by $\{\beta_1, \dots, \beta_k\}$, which is contained, up to homotopy, in both $\Sigma_i$ and hence in $\Sigma_i^\circ$. Therefore, $\gamma \subset \Sigma$, up to homotopy.

(2) Let $\gamma \subset \Sigma$. Then $\gamma$ is up to homotopy contained in both $\Sigma_i$. Since each $\Sigma_i$ is spanned by $\mathcal{C}$, $\gamma$ is a concatenation of curves in $\mathcal{C}$. □

Note that uniqueness requires the minimality assumption, even for finite collections of curves on finite-type surfaces. For example, in Figure 5, let $\mathcal{C}$ be a pair of curves $\alpha$ and $\beta$ that intersect once. Then a one-holed torus with a point removed from the boundary satisfies conditions (1) and (2), but it is not minimal. The subsurface spanned by $\mathcal{C}$ is a one-holed torus (with compact boundary).

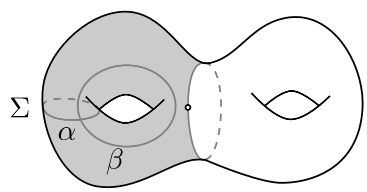


FIGURE 5. A (non-minimal) subsurface $\Sigma$ whose curves are concatenations of $\alpha$ and $\beta$.

If $\mathcal{C}$ is an infinite collection of curves, then a subsurface spanned by $\mathcal{C}$ doesn't always exist — see for instance Figure 6. A simple condition that ensures the existence of a spanned subsurface is that $\mathcal{C}$ be *locally finite*, meaning that every compact set of $S$ intersects only finitely many curves in $\mathcal{C}$. The main goal of this section is to give a condition on $\mathcal{C}$ that guarantees it spans a subsurface. In fact, under the additional assumption that $\mathcal{C}$ is closed under concatenation, we give a necessary and sufficient condition for a collection $\mathcal{C}$ to span a subsurface (Proposition 5.5). This assumption holds for some natural collections of curves, such as curves which are periodic or wandering with respect to a mapping class:

**Lemma 5.2.** *Let $f$ be a mapping class on $S$. Every curve obtained as a concatenation of $f$–periodic curves is periodic, and every curve obtained as a concatenation of wandering curves is wandering.*

*Proof.* We first consider periodic curves. A curve is a concatenation of curves $\gamma_1, \dots, \gamma_n$ if and only if it is contained in the subsurface $F$ spanned by $\{\gamma_1, \dots, \gamma_n\}$. Note that $F$ is connected and has negative Euler characteristic. Since each $\gamma_i$ is $f$–periodic, we can take a suitable power so that $f^k$ fixes $\gamma_i$ for all $i$. Since $F$ is spanned by $\{\gamma_i\}$, $f^k(F)$ is spanned by $\{f^k(\gamma_i)\} = \{\gamma_i\}$, so $f^k(F)$ is isotopic to $F$. Since $F$ has negative Euler characteristic, $f^k$ restricted to $F$ is isotopic to a periodic map of $F$ (see Lemma 2.5), so every curve in $F$ is $f$–periodic.

The proof for wandering curves is similar. Let $F$ be the subsurface spanned by a finite collection $\{\gamma_1, \dots, \gamma_n\}$ of wandering curves. It suffices to show that for all compact subsurface $K \subset S$, $f^{\pm k}(F)$ can be homotoped to be disjoint from $K$ for all sufficiently large $k$. Since each $\gamma_i$ is wandering, we can find $k_0$ such that $f^{\pm k}(\gamma_i)^* \cap K = \emptyset$ for all $k \geq k_0$ and all $i$. Then $f^{\pm k}(F)$ is spanned by $\{f^{\pm k}(\gamma_i)\}$, so $f^{\pm k}(F)$ can be homotoped away from $K$. $\square$

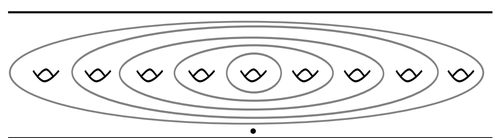

FIGURE 6. The beginning of a collection of curves not spanning a subsurface.

As mentioned before, if $\mathcal{C}$ is a finite collection of curves then there is an explicit way to construct the subsurface spanned by $\mathcal{C}$. If $\mathcal{C}$ is any collection of curves, a first attempt to construct a candidate for the subsurface spanned by $\mathcal{C}$ would be: enumerate the curves in $\mathcal{C}$, take the almost geodesic subsurface spanned by the first $n$ curves and then take the union of these subsurfaces. However, this union need not be the right candidate if there are annular components or homotopic boundary components. For instance, let $\alpha$ be a curve on a surface $S$ and consider the collection $\mathcal{C}$ of all curves disjoint from $\alpha$; the subsurface spanned by $\mathcal{C}$ is the complement of an annulus about $\alpha$, but the naive procedure yields the whole surface $S$. So we need a more complicated procedure to deal with this issue.

is a little confusing. do we get the whole ace? Also, do we not de $\alpha$? –jt

Recall the sets $\mathcal{C}_{\text{is}}, \mathcal{C}_{\text{int}}$ and $\mathcal{C}_{\text{a}}$ defined before:

$$\mathcal{C}_{\text{is}} := \{\gamma \in \mathcal{C} \mid \forall \gamma' \in \mathcal{C},\ i(\gamma, \gamma') = 0\}$$
$$\mathcal{C}_{\text{int}} := \mathcal{C} \smallsetminus \mathcal{C}_{\text{is}}$$
$$\mathcal{C}_{\text{a}} := \{\gamma \in \mathcal{C}_{\text{is}} \mid \gamma \text{ is not a concatenation of curves in } \mathcal{C}_{\text{int}}\}$$

Enumerate the curves in $\mathcal{C}_{\text{int}} = \{\gamma_0, \gamma_1, \dots\}$. Up to reordering, we can assume that $i(\gamma_0, \gamma_1) \neq 0$. Then for every $i \geq 1$, let

$$\mathcal{C}^i_{\text{int}} := \{\gamma_j \mid j \leq i \text{ and } \exists\, j' \leq i \text{ such that } i(\gamma_j, \gamma_{j'}) \neq 0\}.$$

Note that $\mathcal{C}^1_{\text{int}} = \{\gamma_0, \gamma_1\}$, $\mathcal{C}^i_{\text{int}} \subset \mathcal{C}^{i+1}_{\text{int}}$, and $\mathcal{C}_{\text{int}} = \bigcup_{i \geq 1} \mathcal{C}^i_{\text{int}}$. Moreover, if $\mathcal{C}^i_{\text{int}} \neq \mathcal{C}^{i+1}_{\text{int}}$, to obtain $\mathcal{C}^{i+1}_{\text{int}}$ we add curves which either intersect curves in $\mathcal{C}^i_{\text{int}}$ or intersect each other. In particular, each component of the subsurface spanned by $\mathcal{C}^i_{\text{int}}$ has negative Euler characteristic.

We now let $F^i = F(\mathcal{C}^i_{\text{int}})$ be the geodesic representative of the *interior* of the subsurface spanned by $\mathcal{C}^i_{\text{int}}$ and define

$$F = F(\mathcal{C}_{\text{int}}) := \bigcup_{i \geq 1} F^i.$$

By construction every connected component of $F^i$ is an open subset (homotopic to a subsurface), and thus $F$ is an open subset of $S$. We will call $F$ the *set spanned* by $\mathcal{C}_{\text{int}}$.

We also define $F(\mathcal{C})$ to be the union

$$F(\mathcal{C}) := F(\mathcal{C}_{\text{int}}) \cup \bigcup_{\alpha \in \mathcal{C}_{\text{a}}} A(\alpha),$$

where, for every $\alpha$, $A(\alpha)$ is a regular neighborhood of the geodesic representative of $\alpha$, chosen so that they are pairwise disjoint and disjoint from $F(\mathcal{C}_{\text{int}})$.

We will call a collection $\mathcal{C}$ of curves *good* if:

(a) the curves in $\mathcal{C}_{\text{a}}$ when realized as geodesic, form a locally finite collection, i.e. they do not accumulate anywhere; and
(b) for every $p \in \partial F$ there is a closed disk $B$ centered at $p$ such that $B \cap F$ is either $B$ minus a diameter, or a connected component of $B$ minus a diameter.

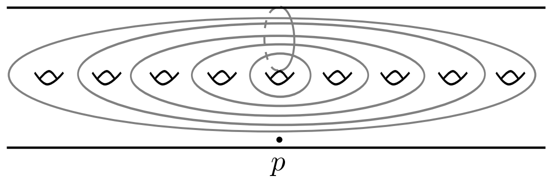


FIGURE 7. A collection of curves which is not good.

For instance, the collection of curves depicted in Figure 6 is not good, because the isolated curves accumulate somewhere. Similarly, the collection in Figure 7 is not good either, because condition (b) is not satisfied at the point $p$ in the drawing (onto which the curves accumulate). In both examples, the collection of curves do not span a subsurface. Indeed, we will show that the condition of being good is equivalent to $\mathcal{C}$ admitting a spanning subsurface (provided $\mathcal{C}$ is closed under concatenation).

Let $K \subset S$ be a finite-type subsurface with compact totally geodesic boundary. Each $F^i$ intersects $K$ in a disjoint union of open subsurfaces and disks with at most one puncture. By a *rectangle* we mean a connected component $R$ of $K \cap F^i$ that is homeomorphic to a closed disk and whose boundary is split into *horizontal sides* lying in $\partial F^i$ and *vertical sides* lying in $\partial K$. All components of $K \cap F^i$ that are not rectangles are called *essential.* Let $K_e^i$ be the union of all the essential components of $K \cap F^i$.

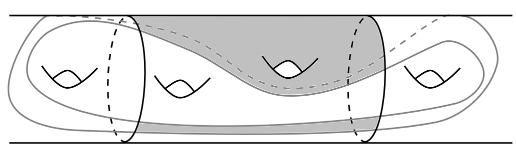

FIGURE 8. A compact subsurface $K$, whose boundary is in black, and the boundary of a subsurface $F^i$ in gray. The shaded part is the intersection, formed by an essential component and a rectangle.

**Lemma 5.3.** *For all $i \geq 2$, $K_e^i \subset K_e^{i+1}$, and for large $i$ it stabilizes. More precisely, there exists an open subsurface $K_e \subset K$ and such that for all sufficiently large $i$, $K_e^i$ is isotopic to $K_e$.*

*Proof.* This follows from Euler characteristic considerations. □

So for any finite-type subsurface $K$ with compact totally geodesic boundary, there exists $I$ such that for all $i \geq I$, the topology of the essential components $K_e^i$ stabilizes, and further increasing $i$ only adds new rectangles or adjoins previous ones. We define:

$$\mathcal{R}(K) = \{R \mid R \subset K \cap F^i \text{ is a rectangle for some } i \geq I \text{ and is not contained in any } K_e^i\}.$$

Two parallel rectangles in $\mathcal{R}(K)$ are declared *equivalent* if they are contained in the same connected component of $F^i \cap K$ for some $i$. The following Lemma is not hard to prove, and we leave it as an exercise.

**Lemma 5.4.** *The relation defined above is an equivalence relation on $\mathcal{R}(K)$. Moreover, suppose $R_1$, $R_2$, $R_3$ are three parallel rectangles with $R_2$ contained in the rectangle between $R_1$ and $R_3$. If $R_1$ and $R_3$ are equivalent, then all three are equivalent.*

This allows us to naturally organize parallel rectangles into equivalence classes. As $i$ increases, the number of equivalence classes may increase. We have the following proposition.

**Proposition 5.5.** *Suppose $\mathcal{C}$ is a collection of curves closed under concatenation. The following statements are equivalent:*

*(1) There is a subsurface spanned by $\mathcal{C}$.*
*(2) The curves in $\mathcal{C}_a$ do not accumulate anywhere and for every compact subsurface $K$ with totally geodesic boundary, the number of equivalence classes in $\mathcal{R}(K)$ is finite.*
*(3) $\mathcal{C}$ is good.*

*Proof.* [(1) $\Rightarrow$ (2)] Let $\Sigma$ be a surface spanned by $\mathcal{C}$. By contradiction, suppose the curves in $\mathcal{C}_{\mathrm{a}}$ accumulate somewhere. Then we can find a sequence of curves $\gamma_j \in \mathcal{C}_{\mathrm{a}}$ and a curve $\alpha$ such that $i(\gamma_j, \alpha) \neq 0$ for all $j$. Since $\Sigma$ is a subsurface, $\Sigma \cap \alpha$ consists of finitely many arcs, with at least one–denoted it $a$–intersecting at least three curves, say $\gamma_{j_1}, \gamma_{j_2}$ and $\gamma_{j_3}$. Assume that $\gamma_{j_2} \cap a$ is between $\gamma_{j_1} \cap a$ and $\gamma_{j_3} \cap a$. Then the pair of pants containing $\gamma_{j_1} \cup a \cup \gamma_{j_3}$ is in $\Sigma$ and contains a curve $\beta$ which intersects $\gamma_{j_2}$ essentially. As $\beta \subset \Sigma$, it is a concatenation of curves in $\mathcal{C}$, and since $\mathcal{C}$ is closed under concatenation we have $\beta \in \mathcal{C}$, contradicting the definition of $\mathcal{C}_{\mathrm{a}}$. This shows that the curves in $\mathcal{C}_{\mathrm{a}}$ cannot accumulate anywhere.

Next, suppose there is a finite-type subsurface $K$ with compact totally geodesic boundary such that $\mathcal{R}(K)$ has infinitely many equivalence classes. Then we can find infinitely many parallel rectangles $R_i$ that are not equivalent. Let $\gamma_i$ be a curve in $F$ passing through $R_i$. By construction, each $\gamma_i$ is a concatenation of subarcs of curves in $\mathcal{C}$, hence $\gamma_i \subset \Sigma$. Since $\Sigma$ is a subsurface, $\Sigma \cap K$ is a finite union of connected components. In particular, there is some component $X$ of $\Sigma \cap K$ containing $\gamma_i \cap K$ for infinitely many $i$. This implies that we can find distinct curves $\gamma_{i_1}$ and $\gamma_{i_2}$ such that the rectangle $R$ between $\gamma_{i_1} \cap K$ and $\gamma_{i_2} \cap K$ is included in $\Sigma$. The boundary of this rectangle is a concatenation of curves in $\mathcal{C}$, which implies that $R$ is contained in $F^l \cap K$ for some $l$, so $R_{i_1}$ and $R_{i_2}$ are equivalent, a contradiction.

[(2) $\Rightarrow$ (3)] We just need to verify property (b) in the definition of "good". By construction, $\partial F$ is a union of limits of simple closed geodesics. We claim that for every finite-type subsurface $K$ with compact totally geodesic boundary, $\partial F \cap K$ has finitely many connected components. Otherwise, as $K$ is of finite type, there are infinitely many components $l_j$ of $\partial F \cap K$ parallel to a given arc from $\partial K$ to $\partial K$. Then there are infinitely many rectangles between the $l_j$ which are not in $F$ and therefore infinitely many rectangles that are not equivalent, a contradiction.

Therefore, for every $p \in \ell \subset \partial F$ we can find a sufficiently small ball $B$ such that $\ell \cap B$ coincided with $\partial F \cap B$ and it is a single arc from $\partial B$ to $\partial B$. In particular, $B \smallsetminus \partial F = B \smallsetminus \ell$ has two components, each of which is either entirely contained in $F$ or in $S \smallsetminus F$. Moreover, since $\ell \subset \partial F$, at least one of the components intersects $F$, and hence is contained in $F$. So $B$ is the required disk.

[(3) $\Rightarrow$ (1)] The condition on boundary points implies that $\partial F$ contains no non-proper component and that its components do not accumulate anywhere. If $\gamma \subset \partial F$ is a compact boundary component, we can find some finite-type subsurface $K$ with compact boundary component such that $\gamma \subset K$. As $K_e^i$ stabilizes, $\gamma$ is eventually a boundary component of some $F^i$, and therefore is an essential curve. If $\gamma \subset \partial F$ is a non-compact boundary component, since it is the limit of simple closed geodesics, it is geodesic, and therefore it is a line. Moreover, since each component of $\partial F$ is a simple closed geodesic or the limit of simple closed geodesics, for any $\ell \subset \partial F$, either

- for every $p \in \ell$ there is a closed disk $B$ centered at $p$ such that $B \cap F$ is a connected component of a disk minus a diameter, or
- for every $p \in \ell$ there is a closed disk $B$ centered at $p$ such that $B \cap F$ is a disk minus a diameter.

Let $T_1$ be the collection of lines and curves in $\partial F$ for which the first condition holds, and $T_2$ be the collection of lines and curve for which the second holds.

For every $\ell \in T_2 \cup \mathcal{C}_{\mathrm{a}}$ we can find a regular neighborhood $N(\ell)$ such that:

- $N(\ell)$ is open if $\ell \in T_2$ and closed if $\ell \in \mathcal{C}_{\mathrm{a}}$;
- $N(\ell)$ is disjoint from $\partial F \cup \mathcal{C}_{\mathrm{a}} \smallsetminus \{\ell\}$;
- if $\ell_1 \neq \ell_2$, then $N(\ell_1) \cap N(\ell_2) = \emptyset$, and
- the neighborhoods don't accumulate anywhere.

We can find such neighborhoods by the properness of components of $\partial F$, together with the non-accumulation assumption on $\mathcal{C}_{\mathrm{a}}$ (we can fix a compact exhaustion of the surface and construct the neighborhoods piece by piece).

Define

$$\Sigma := \left( F \cup \bigcup_{\ell \in T_1} \ell \cup \bigcup_{\gamma \in \mathcal{C}_{\mathrm{a}}} N(\gamma) \right) \smallsetminus \bigcup_{\ell \in T_2} N(\ell).$$

Then $\Sigma$ is a two-dimensional submanifold. Its boundary components are either homotopic to boundary components of $F$ (hence are essential curves or lines), or homotopic to curves in $\mathcal{C}_{\mathrm{a}}$ (hence are essential curves). Therefore, $\Sigma$ is a subsurface.

It is clear that $\Sigma$ contains all curves in $\mathcal{C}$. Moreover, if $\gamma$ is a curve in $\Sigma$, either it is contained in an annular component, and therefore is in $\mathcal{C}$, or it is homotopic into some $F^i$ (by the same argument for compact components of $\partial F$), and therefore $\gamma$ is a concatenation of curves in $\mathcal{C}$. Finally, $\Sigma$ is properly homotopic to $F \cup \bigcup_{\gamma \in \mathcal{C}_{\mathrm{a}}} N(\gamma)$. Since any surface containing all curves in $\mathcal{C}$ must contain $F \cup \bigcup_{\gamma \in \mathcal{C}_{\mathrm{a}}} \gamma$ up to proper isotopy, $\Sigma$ satisfies the minimality hypothesis, and hence is a subsurface spanned by $\mathcal{C}$. □

**Corollary 5.6.** *If $X$ and $Y$ are subsurfaces spanned by two collections of curves $\mathcal{C}_X$ and $\mathcal{C}_Y$ and $i(\alpha, \beta) = 0$ for every $\alpha \in \mathcal{C}_X, \beta \in \mathcal{C}_Y$, then $X$ and $Y$ are disjoint.*

*Proof.* This follows from the explicit construction of a subsurface spanned by a set of curves given in the proof of Proposition 5.5, together with the uniqueness of spanned subsurfaces (Lemma 5.1). □

We record a consequence of having a non-good collection of curves. This will be our main tool for showing certain collections of curves are good.

**Lemma 5.7.** *Let $\mathcal{C}$ be a collection of curves closed under finite concatenations. Suppose $\mathcal{C}$ is not good. Then there is a simple closed geodesic $\alpha$ and two infinite sequences of simple closed geodesics $\beta_j, \gamma_j$, $j \in \mathbb{N}$, with the following properties:*

*(1) $\alpha, \gamma_j \notin \mathcal{C}$ but $\beta_j \in \mathcal{C}$, for all $j$.*
*(2) every $\beta_j$ intersects $\alpha$ and the $\beta_j$'s are pairwise disjoint.*

(3) *For every $j$, there is a subarc $\tau_j \subset \alpha$ joining $\beta_j$ to $\beta_{j+1}$ so that the regular neighborhood of the union $\tau_j \cup \beta_j \cup \beta_{j+1}$ is a pair of pants whose third boundary component is $\gamma_j$. Furthermore, $\tau_j$'s are pairwise disjoint and $\tau_j$ is disjoint from $\beta_k$ for $k < j$.*

*Proof.* By Proposition 5.5, either there is a compact subsurface $K$ with totally geodesic boundary such that $\mathcal{R}(K)$ contains infinitely many equivalence classes, or the curves in $\mathcal{C}_{\mathrm{a}}$ accumulate somewhere.

In the first case, we can find a sequence $R_i \in \mathcal{R}(K)$ which are all parallel and pairwise not equivalent. Let $\alpha$ be a boundary curve of $K$ containing a vertical side of each $R_i$. Up to passing to a subsequence, we can assume that there is an oriented arc $[p,q] \subset \alpha$ and a sequence of points $v_i \in R_i \cap [p,q]$ converging monotonically to $q$. For every $i$ we can moreover find a curve $\delta_i \in \mathcal{C}$ which is a boundary component of some surface $F^{l_i}$ and intersecting $K$ in a collection of arcs that includes a horizontal arc of $R_i$. This boundary curve is in $\mathcal{C}$ since the collection is closed under concatenation. In this way all $\delta_i$'s are pairwise disjoint since the $F^{l_i}$'s are nested.

After a subsequence, we can assume that the intersections $R_i \cap \alpha$ converge to a point $q \in \alpha$ and that the $\delta_i$'s don't contain $q$. After a further subsequence there will be an oriented arc $[p,q] \subset \alpha$ that contains a vertical side of each $R_i$ and the convergence to $q$ is monotonic along $[p,q]$. Further, any arc of any $\delta_i \cap K$ that intersects $[p,q]$ is contained in some rectangle. Inductively, we can construct a subsequence $\delta_{i_j}$ such that

(i) there is a strand of $\delta_{i_{j+1}}$ below (i.e. closer to $q$) all strands of $\delta_{i_j}$, and
(ii) between the lowest strand of $\delta_{i_j}$ and the highest strand of $\delta_{i_{j+1}}$ which is still below $\delta_{i_j}$ there are at least two other rectangles.

ink changing it like is enough. Am I ? –ff

Let $\tau_j$ be the subarc of $[p,q]$ connecting the two strands of $\delta_{i_j}$ and $\delta_{i_{j+1}}$ in (ii). Let $\gamma_j$ be the third boundary component of a small regular neighborhood of $\delta_{i_j} \cup \tau_j \cup \delta_{i_{j+1}}$. Since $\tau_j$ intersects another $\delta_k$, $\gamma_j$ is essential. Moreover, if $\gamma_j \in \mathcal{C}$, then the rectangles associated with the two strands of $\delta_k$'s would be equivalent, which is a contradiction. Now rename $\beta_j := \delta_{i_j}$.

If the curves in $\mathcal{C}_{\mathrm{a}}$ accumulate somewhere, we can find a curve $\alpha$ intersecting infinitely many $\delta_i \subset \mathcal{C}_{\mathrm{a}}$. As before, up to passing to a subsequence, we can find an arc $[p,q] \subset \alpha$, with some $v_i \in \delta_i \cap [p,q]$ converging to $q$ monotonically. We then define $\delta_{i_j}$, $\gamma_j$ and $\tau_j$ as above. Since each $\gamma_j$ intersects a curve in $\mathcal{C}_{\mathrm{a}}$, $\gamma_j \notin \mathcal{C}$.

Note that in both cases there may be strands of $\beta_k$ that intersect $\tau_j$ when $k > j+1$. $\square$

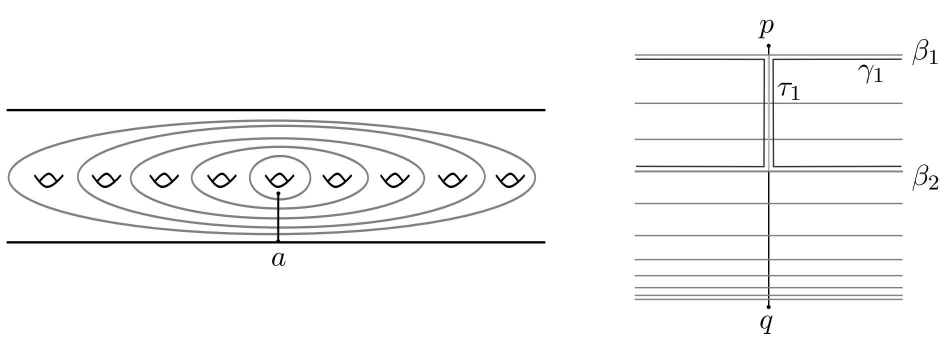


FIGURE 9. On the left-hand side, the arc $a = [p,q]$ for the non-good collection of curves of Figure 6; on the right-hand side, the general situation in the proof of Lemma 5.7.

*Remark* 5.8. In the situation of the previous lemma, and given a homeomorphism $f$, fix an index $j$. Look at $f^n(\alpha)$, $f^n(\beta_j)$ and $f^n(\beta_{j+1})$ and consider an ambient isotopy which sends them to their geodesic representatives. The image through this isotopy of $f^n(\tau_j)$ is a geodesic arc of $f^n(\alpha)^*$ that we denote by $\tau_j^n$. In particular, $f^n(\gamma_j)^*$ is homotopic to $f^n(\beta_j)^* \cup \tau_j^n \cup f^n(\beta_{j+1})^*$.

## 6. Diagonal closure of curves and lines

In this section we discuss two subsurfaces associated to a finite collection of curves and lines: the subsurface spanned by the collection and the diagonal closure of the collection.

Given a finite collection of lines and curves $L$, the subsurface *spanned* by $\mathcal{L}$, denoted $\langle L\rangle$, is the subsurface obtained by putting the curves and lines in minimal position, taking a regular neighborhood of their union, and then filling in all complementary disks with at most one puncture.

Given two rays $r_1$ and $r_2$, we say that they *cobound a half-strip* if there is a proper embedding $\varphi : \mathbb{R}_{\geq 0} \times [0,1] \to S$ such that $\varphi(\mathbb{R}_{\geq 0} \times \{0\}) = r_1$ and $\varphi(\mathbb{R}_{\geq 0} \times \{1\}) = r_2$. We say that two lines $\ell_1$ and $\ell_2$ are *asymptotic* in some direction if they contain rays $r_1 \subset \ell_1$ and $r_2 \subset \ell_2$ which cobound a half-strip.

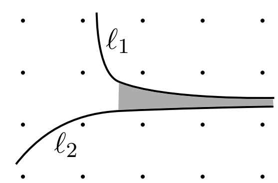


Figure 10. Two asymptotic lines with the half-strip cobounded by two rays.

Let $L$ be a finite collection of curves or proper lines such two elements of $L$ intersect only finitely many times, and no line in $L$ ends in an isolated planar end. We denote by $\langle\langle L\rangle\rangle$ the subsurface of $S$ obtained as follows. First put the curves and lines in $L$ in minimal position. Next form the subsurface $\langle L\rangle$ spanned by $L$. Then, for every two (not necessarily distinct) lines and every direction in which they are asymptotic, adjoin to $\langle L\rangle$ the corresponding half-strip cobounded by the asymptotic rays. Finally fill in any additional complementary disks with at most one puncture. We call $\langle\langle L\rangle\rangle$ the *diagonal closure* of $L$. The following lemma gives a characterization of $\langle\langle L\rangle\rangle$.

**Lemma 6.1.** *Let $L = \{\ell_1, \dots, \ell_N\}$ be a finite collection of curves or lines such that no element end in an isolated planar end of $S$, and any two elements of $L$ intersect only finitely many times. Then for any line $\ell$ that does not end in an isolated planar end, the following statements are equivalent.*

*(1) $\ell$ can be homotoped into $\langle\langle L\rangle\rangle$.*
*(2) For any curve $\alpha$, if $i(\alpha,\ell) \neq 0$, then $i(\alpha,\ell_j) \neq 0$ for some $j$.*
*(3) For every compact subsurface $K$, $\ell$ can be homotoped to $\ell'$ such that $\ell' \cap K \subset \langle L\rangle$.*

*Moreover there is a finite union of curves $\mu$ such that if $\ell$ is a line in $\langle\langle L\rangle\rangle$, then $i(\mu,\ell) \neq 0$.*

*Proof.* We first prove the equivalence of the three conditions.

[(1) $\Rightarrow$ (3)] Since $\langle\langle L\rangle\rangle$ is from $\langle L\rangle$ by adjoining finitely many half-strips and disks, we can first homotope $\ell$ away from all the adjoined disks. For any compact subsurface $K$, there is a homotopy that pushes all the half-strips off of $K$. Let $\ell'$ be the image of $\ell$ under this homotopy. Then $\ell' \cap K \subset \langle L\rangle$.

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[(3) $\Rightarrow$(2)] Let $\alpha$ be a curve with $i(\alpha,\ell) \neq 0$. Let $K$ be an annular neighborhood about $\alpha$ and $\ell'$ a line homotopic to $\ell$ so that $\ell' \cap K \subset \langle L \rangle$. Since $\ell'$ intersects $\alpha$ essentially, $\alpha$ crosses $\langle L \rangle$ essentially. Therefore, there is some $j$ such that $i(\alpha,\ell_j) \neq 0$.

[(2) $\Rightarrow$ (1)] We prove the contrapositive: if $\ell$ cannot be homotoped into $\langle\!\langle L \rangle\!\rangle$, then there is some curve $\alpha$ that intersects $\ell$ but disjoint from all $\ell_j$'s.

Assume all lines are geodesic and $\ell$ is in minimal position with respect to $\langle\!\langle L \rangle\!\rangle$. Note that $\langle\!\langle L \rangle\!\rangle$ is the union of a compact surface with finitely many half-strips. In particular the boundary of $\langle\!\langle L \rangle\!\rangle$ is a finite union of circles and lines. The same therefore holds for $\Sigma := \overline{S \smallsetminus \langle\!\langle L \rangle\!\rangle}$. Moreover, $\Sigma$contains no pairs of boundary rays cobounding a half-strip in $S$, since any such half-strip would be included in the construction of $\langle\!\langle L \rangle\!\rangle$.

Let $a$ be a component of $\ell \cap \Sigma$ and $F$ the component of $\Sigma$ containing $a$. If $a$ is nonseparating in $F$, we can find a curve contained in $F$ intersecting $a$ once and we are done. If $a$ is separating and both components of $F \smallsetminus a$ are not contractible, choose two non-contractible curves $\alpha_1$ and $\alpha_2$ (which may be homotopic to a puncture), one in each component of $F \smallsetminus a$, and a simple arc $b$ connecting them. The regular neighborhood of $\alpha_1 \cup b \cup \alpha_2$ is a pair of pants whose third boundary component gives the required curve $\alpha$. Thus we may assume that $a$ is separating and at least one component $C$ of $F \smallsetminus a$ is contractible. By the classification of contractible surfaces with boundary, $C \cup a$ is a closed disk with points removed from its boundary. Since $\partial\Sigma$ has only finitely many components, $C \cup a$ is in fact a closed disk with finitely many points removed from its boundary. As $\ell$ cannot be homotoped into $\langle\!\langle L \rangle\!\rangle$, one such point is not an end of $a$. It follows that $\Sigma$ contains a half-strip cobounded by two boundary rays, a contradiction (see Figure 11). This finishes the proof of this case.

Finally, we prove the last statement. Write

$$\langle\!\langle L \rangle\!\rangle = K \sqcup S_1 \sqcup \cdots \sqcup S_k,$$

where $K$ is a compact surface and the $S_i$ are pairwise disjoint half-strips. For every $j = 1, \ldots, k$, choose a ray $r_j$ going through $S_j$. As no line ends in a puncture, there is a sequence of essential separating curves, pairwise not homotopic, converging to the end of $r_j$ contained in $S_j$. Choose one such curve $\alpha_j$ sufficiently far out so that $\alpha_j$ intersects the ray $r_j$ and is disjoint from $K$. Let

$$\mu = \bigcup_{j=1}^{k} \alpha_j.$$

If $\ell$ is a line in $\langle\!\langle L \rangle\!\rangle$, then by properness $\ell$ has to exit $K$ along some ray $r_j$, hence it intersects the corresponding $\alpha_j$. This shows $i(\mu,\ell) \neq 0$ as desired. $\square$

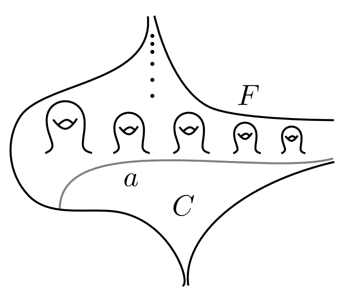


FIGURE 11. The situation in the proof of Lemma 6.1.

*Remark* 6.2. At first one might think that condition (2) in Lemma 6.1 is equivalent to $\ell$ being in $\langle L\rangle$ (up to proper homotopy). The example depicted in Figure 12 shows that this is not the case: condition (2) holds, but $\ell$ cannot be properly homotoped into $\langle L\rangle$.

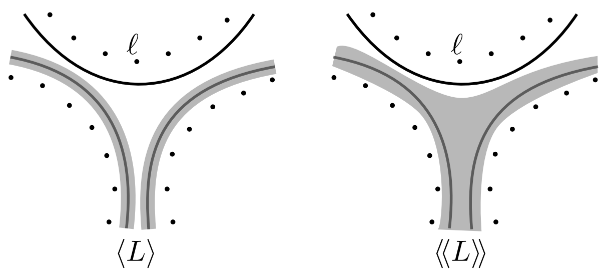


FIGURE 12. $L$ is the union of the gray green lines and $\ell$ is the black one. Then $\ell$ is not properly homotopic into $\langle L\rangle$, but it is in $\langle\langle L\rangle\rangle$.

The reason why we are interested in Lemma 6.1 is the following relationship between $\langle\langle\cdot\rangle\rangle$ and limits of concatenations of curves.

**Lemma 6.3.** *Let $f$ be a well tempered homeomorphism of a surface $S$, and let $\alpha_1,\dots,\alpha_k$ be curves. If $\beta$ is a finite concatenation of $\alpha_1,\dots,\alpha_k$, then*

$$\mathcal{L}^{\pm}(\beta)\subset\Big\langle\Big\langle\bigcup_{i=1}^{n}\mathcal{L}^{\pm}(\alpha_i)\Big\rangle\Big\rangle.$$

*Proof.* Let $\ell\in\mathcal{L}^+(\beta)$ and $\gamma$ a curve intersecting $\ell$. By Lemma 3.6 there are infinitely many $n\geq 0$, such that $i(\gamma,f^n(\beta))\neq 0$. Since $\beta$ is a finite concatenation of the $\alpha_i$'s, there is some $j\in\{1,\dots,n\}$ and infinitely many $n\geq 0$ such that $i(f^n(\alpha_j),\gamma)\neq 0$. Applying Lemma 3.6 again, we obtain some $\ell'\in\mathcal{L}^+(\alpha_j)$ with $i(\ell',\gamma)\neq 0$. Since $\gamma$ was arbitrary, Lemma 6.1 implies that $\ell\subset\langle\langle\bigcup_{i=1}^{n}\mathcal{L}^{\pm}(\alpha_i)\rangle\rangle$ up to proper homotopy. The same argument applies to $\mathcal{L}^-(\beta)$. $\square$

We end this section with a variant of Lemma 6.1.

**Lemma 6.4.** *Let $A\subset B$ be two finite collections of curves or lines such that any two elements intersect only finitely many times. Then there exist finitely many curves $\delta_1,\cdots,\delta_s$ disjoint from lines in $A$, with the following property: if a line $\ell\subset\langle\langle B\rangle\rangle$ cannot be homotoped into $\langle\langle A\rangle\rangle$, then $i(\ell,\delta_i)\neq 0$ for some $i$.*

*Proof.* As in the proof of Lemma 6.1, we can write $\langle\langle B\rangle\rangle$ as the union of a finite-type subsurface $K$ with compact boundary and pairwise disjoint half-strips $S_1,\dots,S_k$. Since $A\subset B$, we can homotope $\langle\langle A\rangle\rangle$ so that it is the union of a finite type subsurface $K'\subset K$ with compact boundary, together with some of the half-strips $S_i$; up to renumbering, say $S_1,\dots,S_l$.

For each $l<j\leq k$, choose a curve $\delta_{j-l}$ associated to the half-strip $S_j$ as in the proof of Lemma 6.1. We append to this collection the set of boundary curves $\delta_{k-l+1},\dots,\delta_s$ of $K'$.

I think we want boun curves of $K'$. –jt

Now suppose a line $\ell$ in $\langle\langle B\rangle\rangle$ cannot be homotoped into $\langle\langle A\rangle\rangle$. If $\ell$ goes through a half-strip not contained in $\langle\langle A\rangle\rangle$, then $\ell$ intersects one of the curves $\delta_i$ with $i\leq k-l$. Otherwise, $\ell\smallsetminus K$ is contained in $\langle\langle A\rangle\rangle$, so $\ell\cap K$ cannot be contained in $K'$ and hence it intersects some $\delta_i$ with $i>k-l$. $\square$

## 7. Well tempered maps: canonical decomposition

By a *decomposition* of $S$ we mean a collection $\{X_i\}$ of pairwise disjoint subsurfaces of $S$ (each of which may be disconnected), such that we can find representatives $\tilde{X}_i$ with $S = \bigcup \tilde{X}_i$ and such that the only overlap between $\tilde{X}_i$ and $\tilde{X}_j$, $i \neq j$, are common boundary components. When $f$ is a homeomorphism of $S$, a decomposition $\{X_i\}$ of $S$ is called *$f$–invariant* if $f(X_i)$ is isotopic to $X_i$ for all $i$. For a subsurface $X \subset S$, we say that $f$ *returns to* $X$ if there is a power $k > 0$ such that $f^k(X)$ is isotopic to $X$. Any such $k > 0$ is called a *return time* of $f$ to $X$. Note that all return times of $f$ to $X$ are multiples of the smallest return time.

Our main proposition in this section is the following.

**Proposition 7.1.** *Suppose $f$ is well tempered. Then there is an $f$–invariant decomposition of $S$ into three subsurfaces, $S_{per}$, $S_\infty$, and $S_0$, with the following properties.*

- *A curve in $S$ is periodic if and only if it can be homotoped into $S_{per}$.*
- *A curve in $S$ is wandering if and only if it can be homotoped into $S_\infty$.*
- *$S_0$ contains no essential, non-peripheral curves.*

*We will call $S_{per}$, $S_\infty$, and $S_0$ the* canonical decomposition *of $f$ on $S$.*

An explicit example of a well tempered map is the map induced by the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ on $S = \mathbb{R}^2 \smallsetminus \mathbb{Z}^2$. In this case, the decomposition is given by four invariant lines, which cobound a subsurface $X = S_0$ homeomorphic to a once-punctured disk with four points removed from the boundary. The rest of the surface is $S_\infty$, consisting of four invariant components (see Figure 13).

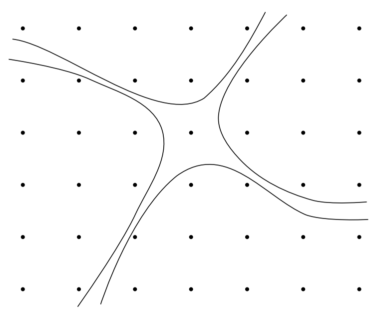

Figure 13. The four invariant lines giving the decomposition of Proposition 7.1 for the map $(x, y) \mapsto (2x + y, x + y)$ on $S = \mathbb{R}^2 \smallsetminus \mathbb{Z}^2$.

For any surface $S$ and any well tempered map $f$, the decomposition is defined as follows. Let $\mathcal{C}_{\mathrm{per}}$ be the collection of $f$–periodic curves, and $\mathcal{C}_\infty$ the collection of $f$–wandering curves. Let $\tilde{S}_{\mathrm{per}}$ be the subsurface spanned by $\mathcal{C}_{\mathrm{per}}$ (whose existence is proved in Lemma 7.4), $S_\infty$ the subsurface spanned by $\mathcal{C}_\infty$ (whose existence is proved in Lemma 7.2).

Fix a hyperbolic metric on $S$. We choose an almost geodesic representative for the components of $\tilde{S}_{\mathrm{per}}$ and $S_\infty$ as follows. For each non-annular component $X$, let $X^*$ be the almost geodesic representative of $X$. Recall this means that we first take the geodesic representative $X^\circ$ of the interior of $X$ and then remove a small regular neighborhood of a boundary component of $X^\circ$ that is homotopic to another boundary component of $X^\circ$. By abuse of notation, we continue to use $\tilde{S}_{\mathrm{per}}$ and $S_\infty$ for the union of the representatives of their components. Define

$$\tilde{S}_0 := \overline{S \smallsetminus (\tilde{S}_{\mathrm{per}} \cup S_\infty)},$$

whose isotopy class is independent of the choice of the hyperbolic metric on $S$. Then:

- $S_{\mathrm{per}}$ is the union of $\tilde{S}_{\mathrm{per}}$ (the subsurface spanned by periodic curves) together with the components of $\tilde{S}_0$ having negative Euler characteristic;
- $S_\infty$ is, as already stated, the subsurface spanned by wandering curves;
- $S_0$ is the union of the components of $\tilde{S}_0$ having non-negative Euler characteristic.

As just seen, a fundamental step in proving Proposition 7.1 is to show that, for a well tempered map, $\mathcal{C}_{\mathrm{per}}$ and $\mathcal{C}_\infty$ span subsurfaces $\tilde{S}_{\mathrm{per}}$ and $S_\infty$. We then need to show that $\tilde{S}_0$ has no interesting topology.

All three proofs proceed by contradiction. Namely, we will show that if the required statement is false, then Lemma 5.7 can be used to produce an essential curve $\alpha$ with infinite limit set $\mathcal{L}(\alpha)$. The arguments for $\tilde{S}_{\mathrm{per}}$ and $S_\infty$ are independent. The proof for $\tilde{S}_0$ is more technical: it relies on the fact that $\tilde{S}_0$ is already a subsurface, being the complement of two subsurfaces, and on the observation that no curve in $\tilde{S}_0$ can be periodic or wandering, hence any curve in $\tilde{S}_0$ must limit onto a non-empty collection of lines.

Let us now prove that $S_\infty$ and $\tilde{S}_{\mathrm{per}}$ are subsurfaces.

**Lemma 7.2.** *If $f$ is well tempered, then $\mathcal{C}_\infty$ spans a subsurface $S_\infty$. Every curve in $S_\infty$ is wandering, and if $\gamma$ is any curve with $\mathcal{L}(\gamma) \neq \emptyset$, then $\mathcal{L}(\gamma)$ does not intersect $S_\infty$ essentially.*

*Proof.* Suppose there is no subsurface spanned by $\mathcal{C}_\infty$. By Lemma 5.2, $\mathcal{C}_\infty$ is closed under concatenation, so we can apply Lemma 5.7 and find curves $\alpha$, $\beta_j$ and $\gamma_j$ and pairwise disjoint arcs $\tau_j \subset \alpha$ as in the lemma. Recall that $\beta_j \in \mathcal{C}_\infty$, but $\alpha, \gamma_j \notin \mathcal{C}_\infty$. In particular, $\mathcal{L}(\gamma_j)$ and $\mathcal{L}(\alpha)$ are nonempty. Since $\gamma_j$ is homotopic to $\beta_j \cup \tau_j \cup \beta_{j+1}$ (and none of these is periodic, by assumption and Corollary 3.9), Lemma 6.3 gives

$$\mathcal{L}(\gamma_j) \subset \langle\!\langle \mathcal{L}(\alpha) \cup \mathcal{L}(\beta_j) \cup \mathcal{L}(\beta_{j+1}) \rangle\!\rangle = \langle\!\langle \mathcal{L}(\alpha) \rangle\!\rangle,$$

where the last equality uses the fact that the $\beta_i$ are wandering, hence $\mathcal{L}(\beta_i) = \emptyset$. Since $\mathcal{L}(\alpha)$ is a finite collection of lines with finite pairwise intersections, Lemma 6.1 provides a finite union of curves $\mu$ such that every line in $\langle\!\langle \mathcal{L}(\alpha) \rangle\!\rangle$ intersects $\mu$. Hence there is a curve $\delta \subset \mu$ that intersects infinitely many $\mathcal{L}(\gamma_j)$. By Lemma 3.7, $\mathcal{L}(\delta)$ intersects infinitely many $\gamma_j$. Since $\mathcal{L}(\delta)$ is finite, some line $\ell \in \mathcal{L}(\delta)$ intersects infinitely many $\gamma_j$.

If $i(\ell, \beta_j) \neq 0$, then $i(\delta, \mathcal{L}(\beta_j)) \neq 0$, which is a contradiction since $\beta_j$ is wandering; the same applies to $\beta_{j+1}$. Therefore $\ell$ must intersect infinitely many of the arcs $\tau_j$, and hence intersects $\alpha$ infinitely many times. This is impossible since $\ell$ is a proper line. This contradiction shows that $\mathcal{C}_\infty$ spans a subsurface $S_\infty$.

For the second statement, if $\gamma$ is a curve such that $\mathcal{L}(\gamma)$ intersects $S_\infty$ essentially, then there exists $\beta \subset S_\infty$ such that $i(\beta, \mathcal{L}(\gamma)) \neq 0$. But every curve in $S_\infty$ is a concatenation of wandering curves, hence $\beta$ is wandering by Lemma 5.2. This contradicts Lemma 3.8. □

We also need the following generalization.

**Lemma 7.3.** *Suppose $f$ is well tempered. Let $L^+$ and $L^-$ be two finite collections of lines with finite pairwise intersections, and such that each of $L^+$ and $L^-$ has finite intersection with every $\mathcal{L}^\pm(\alpha)$. Define*

$$\mathcal{C}_L := \{\beta \mid \mathcal{L}^\pm(\beta) \subseteq \langle\!\langle L^\pm \rangle\!\rangle\}.$$

*Then $\mathcal{C}_L$ spans a subsurface $S_L$ of $S$.*

*Proof.* Note that $\mathcal{C}_L$ contains all wandering curves, and is closed under concatenation by Lemma 6.3.

Assume that $\mathcal{C}_L$ does not span a subsurface. Then we have curves $\alpha, \beta_j, \gamma_j$ and arcs $\tau_j \subset \alpha$ as in Lemma 5.7, where $\beta_j \in \mathcal{C}_L$; $\alpha, \gamma_j \notin \mathcal{C}_L$; and $\gamma_j$ is homotopic to $\beta_j \cup \gamma_j \cup \beta_{j+1}$. In this situation,

$$\mathcal{L}^{\pm}(\gamma_j) \subset \langle\langle \mathcal{L}^{\pm}(\alpha) \cup \mathcal{L}^{\pm}(\beta_j) \cup \mathcal{L}^{\pm}(\beta_{j+1}) \rangle\rangle \subseteq \langle\langle \mathcal{L}^{\pm}(\alpha) \cup L^{\pm} \rangle\rangle.$$

Let $\{\delta_l^+\}_l$ and $\{\delta_m^-\}_m$ be the curves from Lemma 6.4 for the collections $L^{\pm} \subset L^{\pm} \cup \mathcal{L}^{\pm}(\alpha)$. After passing to a subsequence and possibly replacing $f$ by $f^{-1}$ and switching $+$ and $-$, we may assume that for every $j$ there is at least one line in $\mathcal{L}^+(\gamma_j)$ that is not homotopic into $\langle\langle L^+ \rangle\rangle$, hence intersects one of the $\{\delta_l^+\}_l$. Passing to a further subsequence, we may assume that the same curve $\delta \in \{\delta_l^+\}$ intersects infinitely many $\mathcal{L}^+(\gamma_j)$. By construction, $\delta$ does not intersect any of $\mathcal{L}^+(\beta_j)$. It follows by Lemma 3.7 that $\mathcal{L}^-(\delta)$ must intersect $\alpha$ infinitely many times, which is impossible since the map is well tempered. □

**Lemma 7.4.** *If $f$ is well tempered, then $\mathcal{C}_{per}$ spans a subsurface $\tilde{S}_{per}$, and every curve in $\tilde{S}_{per}$ is periodic.*

*Proof.* By Lemma 5.2, $\mathcal{C}_{\text{per}}$ is closed under concatenation. Suppose, for contradiction, there is no subsurface spanned by $\mathcal{C}_{\text{per}}$. Then we can find curves $\alpha$, $\beta_j$ and $\gamma_j$ and arcs $\tau_j \subset \alpha$ as in Lemma 5.7. Recall that $\beta_i \in \mathcal{C}_{\text{per}}$, but $\alpha, \gamma_j \notin \mathcal{C}_{\text{per}}$. Moreover, since $\alpha$ intersects periodic curves, it is not wandering either. Thus $\mathcal{L}(\alpha)$ is non-empty and consists of lines.

Each line in $\mathcal{L}(\alpha)$ is the limit of a uniformly bounded number of strands of $f^{n_i}(\alpha)^*$. Let $N$ be the cardinality of $\mathcal{L}(\alpha)$, counted with multiplicity, and set $J = N + 1$. Consider the finite collection of curves $\gamma_1, \ldots, \gamma_J$. By replacing $f$ by a power, we may assume $f(\beta_j) = \beta_j$ for all $j \leq J + 1$.

Recall that $\gamma_j$ is homotopic to $\beta_j \cup \tau_j \cup \beta_{j+1}$. For each $n$, let $\tau_j^n$ be the geodesic subarc of $f^n(\alpha)^*$ which together with $f^n(\beta_j)^* = \beta_j$ and $f^n(\beta_{j+1})^* = \beta_{j+1}$ is homotopic to $f^n(\gamma_j)^*$ (see Remark 5.8). Since $\gamma_j$ is not periodic, the arcs $\tau_j^n$ cannot be periodic. As there are only finitely many isotopy classes of arcs of bounded length between $\beta_j$ and $\beta_{j+1}$, the length of $\tau_j^n$ cannot be bounded as $n \to \infty$. i

Moreover, the angles of intersections of $\tau_j^n$ with $\beta_j$ and $\beta_{j+1}$ are bounded away from zero; otherwise some sequence of iterates of $\alpha$ would wrap around $\beta_j$ or $\beta_{j+1}$ more and more times, contradicting the tempered condition. Thus any convergent subsequence of $\tau_j^n$ contains a ray starting at $\beta_j$ and a ray starting at $\beta_{j+1}$. By construction, each such ray is contained in a line of $\mathcal{L}(\alpha)$.

Choose a subsequence $f^{n_i}$ so that each (oriented) $\tau_j^{n_i}$ converges. Then the union of all these limit sets contains at least $2J = 2N + 2$ rays. If a collection of $k$ such limiting rays is nested within the same line in $\mathcal{L}(\alpha)$, then the multiplicity of that line is at least $k$. Taking into account both orientations of each line, we see that there can be at most $2N$ rays, a contradiction. This shows the existence of $\tilde{S}_{\text{per}}$. □

**Lemma 7.5.** *If $f$ is well tempered, then $\tilde{S}_{per}$ and $S_\infty$ are disjoint subsurfaces.*

*Proof.* Since they are subsurfaces spanned by curves, if they are not disjoint, by Corollary 5.6 there are curves $\alpha$ in $\tilde{S}_{\text{per}}$ and $\beta$ in $S_\infty$ with $i(\alpha, \beta) > 0$, contradicting Corollary 3.9. □

**Lemma 7.6.** *If $f$ is well tempered, $\tilde{S}_0 := \overline{S \setminus (\tilde{S}_{per} \cup S_\infty)}$ has no essential, non-peripheral curves.*

*Proof.* Suppose, by contradiction, that there is an essential, non-peripheral curve $\delta$ in $\tilde{S}_0$. Then $\delta$ is neither wandering or periodic, so $L := \mathcal{L}(\delta)$ is a nonempty collection of lines. Let $L^{\pm} := \mathcal{L}^{\pm}(\delta)$ and let $S_L$ be the subsurface from Lemma 7.3 spanned by the collection of curves

$\{\gamma\}$ with $\mathcal{L}^{\pm}(\gamma) \subset \langle\!\langle L^{\pm} \rangle\!\rangle$. Put $S_L$ and $\tilde{S}_0$ in minimal position and let $\tilde{S}_L = \tilde{S}_0 \cap S_L$. Note that $S_\infty \subseteq S_L$ and $\tilde{S}_{\text{per}} \cap S_L$ is a compact surface, so the boundaries of $S_L$ and $\tilde{S}_0$ intersect in only finitely many points.

Up to replacing $f$ by its inverse, we may assume $L^+$ is nonempty. By Lemma 6.1, there exists a finite collection of curves $\{\delta_1, \dots, \delta_k\}$ intersecting all lines in $\langle\!\langle L^+ \rangle\!\rangle$. Then Lemma 3.7 implies that for every curve $\gamma$ in $S_L$, either

$$\text{(a) } \mathcal{L}^+(\gamma) = \emptyset, \quad \text{or} \quad \text{(b) } i(\gamma, \mathcal{L}^-(\delta_i)) \neq 0 \text{ for some } i.$$

*Claim:* There exists an essential, nonperipheral curve in $\tilde{S}_L$ satisfying (a).

Indeed, otherwise the finite collection of curves and lines in $\bigcup_i \mathcal{L}^-(\delta_i)$ cuts every curve in $\tilde{S}_L$. Since there are only finitely many intersection points between these curves and lines , it follows that each component of $\tilde{S}_L$ has finite type. Furthermore, since $\tilde{S}_L$ is $f$-invariant, if one of its components is not periodic, then its orbit is a subsurface of $\tilde{S}_L$. In particular the orbit of the component cannot accumulate anywhere; that is, it is wandering. Now consider the component that contains $\delta$. If it is wandering, $\delta$ is wandering, which is impossible. Hence this component must be periodic. But then the first return map is a tempered mapping class of a finite-type surface; in particular, it is periodic and so $\delta$ is periodic as well, which is again impossible. This proves the claim.

why is this true? –jt

By the claim, there exists an essential, nonperipheral curve $\delta'$ in $\tilde{S}_0$ that is forward wandering. Replace $f$ by $f^{-1}$ and run the whole construction on $\delta'$, obtaining the subsurfaces $S_{L'}$ and $\tilde{S}_{L'}$, where $L' = L^{\pm}(\delta')$. Now, every curve $\gamma$ in $S_{L'}$ is $f$-forward wandering (since $\mathcal{L}^+(\gamma) \subset \langle\!\langle \mathcal{L}^+(\delta') \rangle\!\rangle = \emptyset$). The above claim produces another essential, nonperipheral curve in $\tilde{S}_{L'}$ which is $f$-backward wandering, and thus wandering, a contradiction. □

A consequence of the lemma is the following.

**Corollary 7.7.** *Any negative Euler characteristic component of $\tilde{S}_0$ is a pair of pants.*

*Proof.* If the Euler characteristic were smaller than $-1$, there would be essential non-peripheral curves. Hence the Euler characteristic is $-1$ . If the component is not a pair of pants, then it is a pair of pants with some points removed from its boundary, which then contains an essential non-peripheral curve, a contradiction. □

7.1. **Examples.** We end this section with some examples of tempered but not well tempered maps for which the conclusions of Proposition 7.1 are false.

*Example* 7.8. This example shows the existence of a tempered map $f$ such that $\mathcal{L}(\alpha)$ is locally finite for every curve $\alpha$, but the collection of periodic curves does not span a subsurface.

Consider the cylinder $\Sigma := S^1 \times \mathbb{R}$ and fix a Cantor set $K \subset S^1$. Let $S$ be the surface obtained as

$$S := \Sigma \smallsetminus \bigcup_{n=-\infty}^{\infty} K \times \{t_n\},$$

where $t_{\pm\infty} = \pm 1$, $t_0 = 0$, $t_n = -1 + \frac{1}{2^{-n}}$ for $n < 0$ and $t_n = 1 - \frac{1}{2^n}$ for $n > 0$ (see Figure 14).

Fix a monotone non-decreasing homeomorphism $s : \mathbb{R} \to \mathbb{R}$ such that

$$\begin{aligned} s(t_{\pm\infty}) &= t_{\pm\infty} \\ s(t_k) &= t_{k+1} \quad \forall k \in \mathbb{Z}. \end{aligned}$$

Then the map $\mathrm{id} \times s : \Sigma \to \Sigma$ induces a tempered homeomorphism $f$ of $S$, which has no wandering curves. The periodic curves are also fixed curves, which are those that can be homotoped to be vertical between height $-1$ and height 1. In particular $\mathcal{C}_{\text{per}}$ does not span a subsurface.

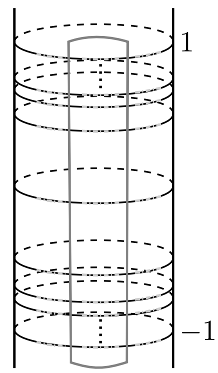


FIGURE 14. A sketch of the surface in Example 7.8, with a periodic curve (in gray) and the Cantor sets in light gray.

*Example* 7.9. This example shows that that there exists a tempered map $f$ such that $\mathcal{L}(\alpha)$ is locally finite for every curve $\alpha$, but $\mathcal{C}_\infty$ does not span a subsurface. It also shows that such an example can be constructed so that a single component of $S_\infty$ is not a subsurface.

In the plane, for any $k \in \mathbb{N}$, let $\ell_k$ be the boundary of

$$[k+1, \infty) \times \left[\frac{1}{k+1} + \varepsilon_k, \frac{1}{k} - \varepsilon_k\right],$$

where $\varepsilon_k = \frac{1}{10k^2}$ (see Figure 15 for a schematic picture). Let $S$ be the plane punctured at the points $\left(m, \frac{1}{n}\right)$, for $n \in \mathbb{N}$ and $m \in \mathbb{Z}$, at the points $(0, m)$, for $m \in \mathbb{Z}$, and for every line $\ell_k$, $k \in \mathbb{N}$, at a sequence of points on the line not accumulating anywhere.

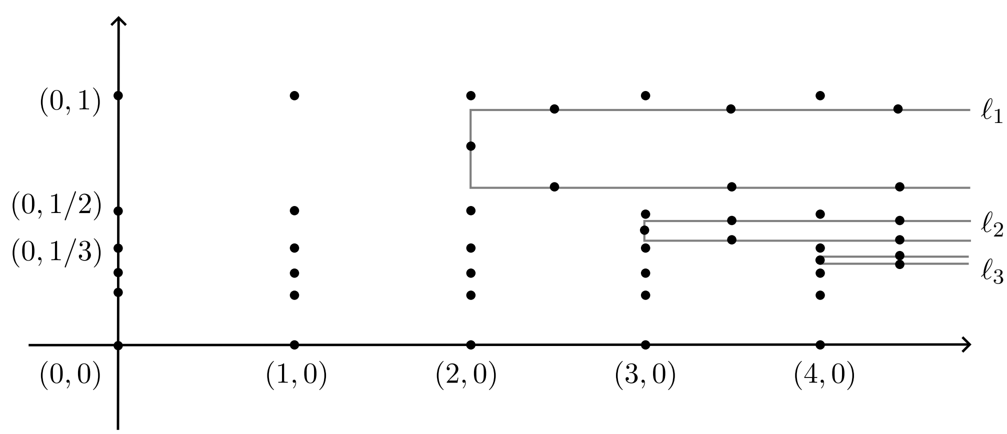


FIGURE 15. The lines $\ell_k$ in the construction of Example 7.9.

The homeomorphism $f$ is defined as follows:

- on a small regular neighborhood (of width less than $\varepsilon_k$) of each horizontal line $y = \frac{1}{n}$ ($n \in \mathbb{N}$) and $y = 0$, the map is induced by $(x, y) \mapsto (x+1, y)$;

- for each $k$, it performs a puncture shift supported on a small regular neighborhood of $\ell_k$, tapered to the identity on the rest of the surface.

One can show that $f$ is tempered. Moreover, $F(\mathcal{C}_\infty)$ is the disjoint union of one strip per line $y = \frac{1}{n}$, $n \in \mathbb{N}$, and of the strips $(k, \infty) \times \left(\frac{1}{k+1} + \frac{\varepsilon_k}{2}, \frac{1}{k} - \frac{\varepsilon_k}{2}\right)$. The closure of each component is a subsurface, but the horizontal strips accumulate onto $y = 0$, so $\mathcal{C}_\infty$ does not span a subsurface. Note moreover that there are no periodic curves.

Finally, we can modify the surface by adding handles connecting consecutive horizontal strips in a translation-invariant way (see Figure 16 for a schematic picture). The induced map on this new surface is still tempered and has no periodic curves. In this case, $F(\mathcal{C}_\infty)$ consists of one component per line $\ell_k$ and a single nonplanar component. This nonplanar component is not homotopic to a subsurface.

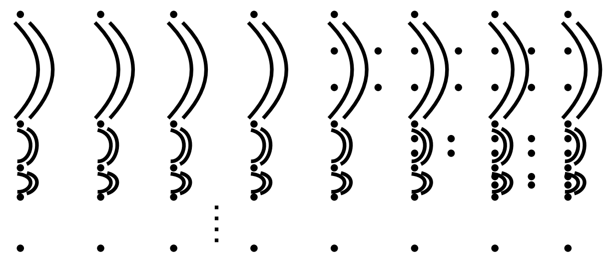

FIGURE 16. The modified surface in Example 7.9.

## 8. Characterization of translations

In this section, we give a characterization of a translation in terms of the dynamics of its action on curves. The main application will be to an well tempered map $f$ and its action on the invariant subsurface $S_\infty$ in its canonical decomposition. Therefore, in this section, we need to work with surfaces with boundary, and we will prove a more general version of Theorem C. Everything in this section is independent of the previous sections.

Extending the notation for curves, when $\delta$ is an essential arc in a hyperbolic surface with totally geodesic boundary, we denote by $\delta^*$ the geodesic arc homotopic to $\delta$ which is perpendicular to the boundary at both endpoints.

**Lemma 8.1.** *Let $X$ be a connected surface, possibly with boundary, which is spanned by its curves. Let $f$ be a homeomorphism of $X$ such that every curve in $X$ is $f$-wandering. Then there are ends $e_\pm$ of $X$, possibly $e_+ = e_-$, such that for every curve $\alpha$ in $X$ $f^{\pm n}(\alpha)^*$ converges to $e_\pm$ as $n \to \pm\infty$.*

We call $e_+$ and $e_-$ the $f$*−attracting* and $f$*−repelling* ends of $X$, respectively.

*Proof.* Let $\alpha$ be a non-peripheral curve in $X$. We first show that $f^n(\alpha)^*$ converges to an end. Let

$$\alpha^* = \alpha_0, \alpha_1, \dots \alpha_k = f(\alpha)^*$$

be a chain of geodesic curves in $X$ connecting $\alpha^*$ and $f(\alpha)^*$, i.e. $i(\alpha_i, \alpha_{i+1}) \neq 0$ for every $i$. For any compact set $K \subset X$, there exists $n_0$ such that for all $n \geq n_0$ and all $i \in \{1, \dots, k\}$, the curves $f^n(\alpha_i)^*$ are disjoint from $K$. Since consecutive ones intersect, they must lie in the same connected component of $X \setminus K$. In particular, $f^n(\alpha)^*$ and $f^{n+1}(\alpha)^*$ lie in the same complementary component of $K$, for all $n \geq n_0$. As this holds for all compact sets of $X$, it follows that $f^n(\alpha)^*$ converges to a single end $e_+$ as $n \to \infty$. The same argument applied to $f^{-1}$ shows that $f^n(\alpha)^*$ converges to an end $e_-$ as $n \to -\infty$.

Now let $\beta$ be another non-peripheral curve in $X$. Choose a chain of curves connecting $\alpha^*$ to $\beta^*$ and repeat the same argument as above to deduce that $f^n(\beta)^* \to e_\pm$ as $n \to \pm\infty$.

Finally, if $\gamma$ is a peripheral curve, then it can be expressed as a concatenation of non-peripheral curves (since $X$ is spanned by its curves), all of which converge under forward (respectively backward) iteration to $e_+$ (respectively $e_-$), and hence so does $\gamma$. □

**Lemma 8.2.** *Let $X$ be a connected surface, possibly with boundary, and let $e_+, e_-$ be two distinct ends of $X$. Then there is a either a curve or a finite collection of disjoint arcs in $X$ separating $e_+$ from $e_-$.*

*Proof.* Let $D(X)$ be the double of $X$ along its boundary (and set $D(X) = X$ if $\partial X = \emptyset$). Let $i : X \hookrightarrow D(X)$ be the inclusion map and $i_* : \mathrm{Ends}(X) \to \mathrm{Ends}(D(X))$ the induced map. Note that $i_*$ is injective: if $e_1 \neq e_2$ are ends of $X$, there is a compact subset $K$ separating them, and therefore $i_*(e_1)$ and $i_*(e_2)$ are separated by the double of $K$ in $D(X)$.

Since $D(X)$ has no boundary, there is a curve $\hat{\delta} \subset D(X)$ separating $i_*(e_+)$ and $i_*(e_-)$. Assume $\hat{\delta}$ is in minimal position with respect to $i(\partial X)$ and let $\delta := i^{-1}(\hat{\delta})$.Then $\delta$ is either a curve in $X$ or a finite collection of pairwise disjoint arcs. To see that $\delta$ separates $e_+$ and $e_-$: if not, there would exist a line $\ell$ in $X$ from $e_+$ to $e_-$ disjoint from $\delta$, and hence a line $i(\ell)$ from $i_*(e_+)$ to $i_*(e_-)$ disjoint from $\hat{\delta}$, contradicting that $\hat{\delta}$ separates the ends. □

**Proposition 8.3.** *Let $X$ be a connected (possibly bordered) surface spanned by its curves. Let $f$ be a homeomorphism of $X$ such that every curve is $f$–wandering but no boundary line is $f$–wandering. Then every essential arc $\gamma$ in $X$ is $f$–wandering.*

confused about peeral curves. Are they of this? Should we something about arcs to them? –jt

our application, we assume that there are peripheral curves, so ould not to complimatters. Otherwise need to add an arent, but that's essen- the same as Case 1: e arc were not wanng, the string of the oon would also have to me longer and longer, e it connects the wanng balloon to a wanng boundary curve. also just seen that e is a commented arent for this case... –ff

*Moreover, assume $X$ is equipped with a complete hyperbolic metric with totally geodesic boundary. If an essential arc $\gamma$ has both endpoints on the same oriented boundary line $\ell$, and if $f$ preserves $\ell$ and its orientation, then both endpoints of $f^n(\gamma)^*$ escape to infinity in the same direction along $\ell$ as $n \to \infty$.*

*Proof.* Fix a complete hyperbolic structure on $X$ with totally geodesic boundary and positive injectivity radius. Recall that for any arc $\gamma$, we denote by $\gamma^*$ its geodesic representative which is orthogonal to the boundary of $X$. We must show that $f^n(\gamma)^*$ leaves every compact set of $X$.

As for curves, denote by $\mathcal{L}^\pm(\gamma)$ the accumulation sets of $\gamma_n^*$ as $n \to \pm\infty$. Applying the same argument as in Lemma 3.7 to arcs, we have that

$$i(\mathcal{L}(\gamma), \beta) \neq 0 \Longleftrightarrow i(\mathcal{L}(\beta), \gamma) \neq 0,$$

where $\beta$ is any curve or arc in $S$. Since curves in $X$ are wandering, $\mathcal{L}(\gamma)$ cannot contain any geodesic segment that meets the interior of $X$ essentially. It follows that the endpoints of $\gamma_n^*$ must escape to infinity along $\partial X$, for otherwise $\mathcal{L}(\gamma)$ would contain an arc or a ray perpendicular to $\partial X$. Hence, if $\mathcal{L}(\gamma) \neq \emptyset$, then it is a union of boundary components.

*Case 1.* Both endpoints of $\gamma$ are on the same oriented boundary line $\ell$ and $f$ preserves the oriented line $\ell$. In this case, we close up $\gamma$ along $\ell$ to obtain a curve $\tilde{\gamma}$. Assume for contradiction that $\mathcal{L}^+(\gamma) \neq \emptyset$, i.e. $\gamma$ is not forward wandering. Then there is a compact set $K \subset X$ and a sequence $n_i \to \infty$ such that $f^{n_i}(\gamma)^* \cap K \neq \emptyset$ for all $i$.

Note first that the distance between the endpoints of $f^{n_i}(\gamma)^*$ goes to 0 as $i \to \infty$. Indeed, otherwise after a further subsequence the geodesic $f^{n_i}(\tilde{\gamma})^*$ would be in a fixed Hausdorff neighborhood of $f^{n_i}(\gamma)^*$ and would intersect a fixed neighborhood of $K$, contrary to the assumption that $\tilde{\gamma}$ is wandering.

Up to homotopy, we can express $f^{n_i}(\gamma)$ as $\tau_{n_i} * f^{n_i}(\tilde{\gamma})^* * \overline{\tau}_{n_i}$ for a geodesic arc $\tau_{n_i}$ perpendicular at the endpoints to $\ell$ and $f^{n_i}(\tilde{\gamma})^*$ respectively. If the lengths of the $\tau_{n_i}$ are uniformly bounded from above, $\tau_{n_i} * f^{n_i}(\tilde{\gamma})^* * \overline{\tau}_{n_i}$ goes to infinity, as $\tilde{\gamma}$ is wandering, and we are done. So we can assume that the lengths of the $\tau_{n_i}$ are bounded from below. Since the injectivity radius is positive, also the length of $f^{n_i}(\tilde{\gamma})$ is bounded from below. Therefore, $f^{n_i}(\gamma)^*$ and $\tau_{n_i} * f^{n_i}(\tilde{\gamma})^* * \overline{\tau}_{n_i}$ will be a uniformly bounded distance from each other. As $f^{n_i}(\gamma)^*$ intersects $K$ for all $i$ and $f^{n_i}(\tilde{\gamma})^*$ goes to infinity, $\tau_{n_i}$ intersect a neighborhood $L$ of $K$, which can be taken to be a finite-type surface with totally geodesic boundary.

Define now an (immersed) curve $\tilde{\delta}$ by concatenating $\gamma * \alpha * f(\gamma) * \beta$, where $\alpha$ and $\beta$ are arcs in $l$. The argument from Lemma 5.2 also shows that immersed curves are wandering. Therefore, as $i \to \infty$, also $f^{n_i}(\tilde{\delta})^*$ go further and further from $L$.

Let $p_i$ be the last intersection of $\tau_{n_i}$ with $L$. If $\tau_{n_i}$ and $\tau_{n_{i+1}}$ are not within a small neighborhood of each other until a large fixed distance past $p_i$ and $p_{i+1}$, then $f^{n_i}(\tilde{\delta})^*$ will be homotopic to a curve of the form $f^{n_i}(\tilde{\gamma}) * \eta * f^{n_i+1}(\tilde{\gamma}) * \overline{\eta}$ for an arc $\eta$ perpendicular to both $f^{n_i}(\tilde{\gamma})$ and $f^{n_i+1}(\tilde{\gamma})$ and with parts of $\tau_{n_i}$ and $\tau_{n_i+1}$ that don't fellow travel contained in a bounded neighborhood of $\eta$. It follows that $f^{n_i}(\tilde{\delta})^*$ intersects a bounded neighborhood of $L$ and this can happen for only finitely many $i$. Thus the initial portions of $\tau_{n_i}$ and $\tau_{n_i+1}$ (until $p_i$ and $p_{i+1}$) together with a small geodesic arc on $\ell$ and on a boundary component of $L$ determine a geodesic quadrilateral which is nullhomotopic. The same argument can now be applied after replacing $n_i$ with $n_{i+1}$, and inductively for all large powers. Thus for large $i$ all longer and longer initial portions of the $\tau_{n_i}$ are all homotopic, keeping the endpoints on $L$ and $\ell$. But then the endpoints on $\ell$ cannot go to $\infty$. This contradiction establishes that $\gamma$ is forward wandering.

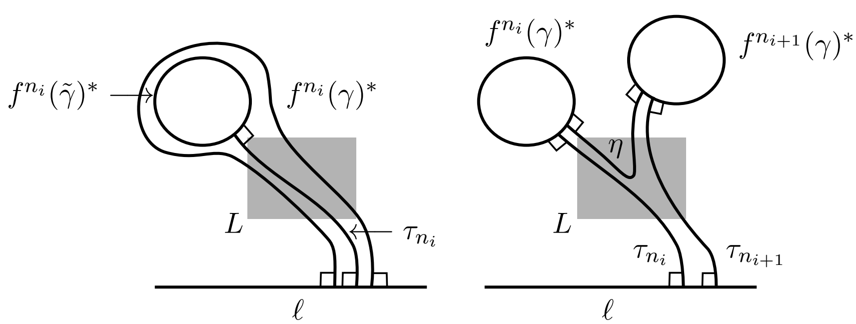


FIGURE 17. Schematic pictures in Case 1

*Case 2.* Both endpoints of $\gamma$ are on a periodic line $\ell$. Then apply Case 1 to the power of $f$ that preserves the oriented line $\ell$.

*General case.* We have seen that the limit set of an arc $\gamma$, if non-empty, can only consist of boundary lines. If $\ell$ is a boundary line in $\mathcal{L}(\gamma)$, let $\beta$ be an essential arc with both endpoints on $\ell$. By the previous cases, $\beta$ is wandering, so by Lemma 3.7

$$i(\mathcal{L}(\beta), \gamma) = 0 \iff i(\beta, \mathcal{L}(\gamma) = 0.$$

However, $\beta$ intersects $\ell \in \mathcal{L}(\gamma)$, a contradiction. □

**Corollary 8.4.** *Let $X$ be a connected (possibly bordered) surface spanned by its curves. Let $f$ be a homeomorphism of $X$ such that every curve is $f$–wandering but no boundary line is* *wandering. Let $D(X)$ be the double of $X$ and let $\hat{f}$ be the induced map on $X$. Then every curve in $D(X)$ is $\hat{f}$-wandering.*

*Proof.* The double $D(X)$ is cut by the collection of all curves in each copy of $X$ and all curves obtained as doubles of essential arcs in $X$. By assumption and by the previous proposition, all these curves are wandering. Then, by Lemma 5.2, every curve in $D(X)$ is wandering. □

**Theorem 8.5.** *Let $X$ be a connected (possibly bordered) surface spanned by its curves. Let $f$ be a homeomorphism of $X$ such that every curve is $f$–wandering but no boundary line is $f$–wandering. Then there is a hyperbolic metric on $X$ and an isometric translation on $X$ isotopic to $f$.*

*Remark* 8.6. If the surface has empty boundary, Theorem 8.5 reduces to Theorem C.

*Proof.* By Lemma 8.1 there are ends $e_+$ and $e_-$ of $X$ to which all curves converge under forward/backward iteration. In particular, $f(e_\pm) = e_\pm$. There are now two cases.

*Case 1: $e_+ \neq e_-$.* Let $\delta$ be as in Lemma 8.2. Choose $n$ so that $i(f^n(\delta), \delta) = 0$. Set $\delta_i = f^{ni}(\delta)^*$, each of which is an multiarc separating $e_+$ and $e_-$. Note that $\delta_{i+1} \subset H_i^+$, where $H_i^+$ is the component of $X \setminus \delta_i$ containing $e_+$. This shows that the dual graph is a line on which $f^n$ acts as a translation. We can homotope $f^n$ so it is a translation on $X$, with the region between $\delta_0$ and $\delta_1$ a fundamental domain for the action of $f^n$.

We now argue that $f$ acts as an isometric translation of $X$. Let $Q = X/\langle f^n \rangle$ be the quotient surface. Since $f$ commutes with $f^n$, $f$ descends to a map on $Q$, whose $n$-th power is the identity map on $Q$, so $f$ is a periodic map of $Q$. By Proposition 2.7, there is a hyperbolic metric on $Q$ for which $f$ acts as an isometry of $Q$. Lifting this metric to $X$ also makes $f$ an isometry of $X$. Since $f^n$ is a translation of $X$, $f$ acts on $X$ as a covering transformation, so it is also a translation of $X$.

*Case 2: $e_+ = e_-$.* Let $D(X)$ be the double of $X$ along its boundary if $\partial X \neq \emptyset$; otherwise $D(X) = X$. By Corollary 8.4, every curve in $D(X)$ is wandering with respect to the map induced by $f$ (which we will also denote by $f$, abusing notation). Let $e$ be the end of $D(X)$ which is the image of $e^\pm$ under the natural map $\mathrm{Ends}(X) \to \mathrm{Ends}(D(X))$. Note that all curves of $D(X)$ converge to $e$ under forward and backward iteration of $f$.

The goal is to construct an exhaustion of $D(X)$ by $f$-invariant subsurfaces, each of which falls into Case 1. If $D(X) \neq X$, we will perform every operation and make every choice below respecting the symmetry given by the doubling.

To construct this exhaustion, fix a curve $\alpha_1$ and choose a sequence of curves

$$\alpha_1, \alpha_2, \ldots, \alpha_k = f(\alpha_1)$$

in $D(X)$ so that $i(\alpha_i, \alpha_{i+1}) > 0$. The union of the orbits of the $\alpha_i$ is a locally finite collection (since they are wandering), so they span a subsurface, which we denote by $T_1$. By construction, $T_1$ is $f$-invariant (up to isotopy) and $\langle f \rangle \simeq \mathbb{Z}$ acts properly discontinuously and cocompactly on $T_1$. Therefore $T_1$ has two ends $t_\pm$ – in fact, it is quasi-isometric to $\mathbb{Z}$ – and every curve in $T_1$ converges under forward (respectively, backward) iteration to $t_+$ (respectively, $t_-$). Modify $f$ by an isotopy so that $f(T_1) = T_1$.

Repeating this for a larger collection of curves yields a sequence of subsurfaces $T_1 \subset T_2 \subset \cdots$ exhausting $D(X)$, each of which is 2-ended and $f$–invariant.

As in Case 1, we can find exponents $n_i$ so that $Q_i := T_i/\langle f^{n_i} \rangle$ is a surface and $f$ induces a finite-order map on each $Q_i$. We may assume that $n_i$ divides $n_{i+1}$ for all $i$. Note that each $Q_i$ is of finite-type, since each $T_i$ is spanned by finitely many orbits of curves. By Proposition 2.7, we can find a hyperbolic structure $Y_1$ on $Q_1$ which is $f$–invariant and thus can be lifted

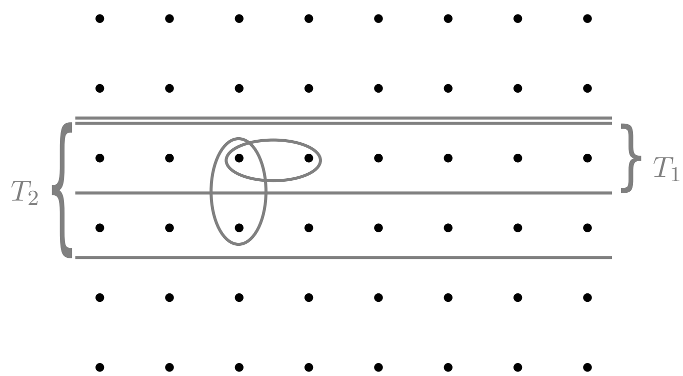


FIGURE 18. An example of the exhaustion $T_1 \subset T_2 \subset \dots$, where the map is the horizontal shift to the right, $T_1$ is the surface spanned by the orbit of the "horizontal" curve and $T_2$ the surface spanned by the orbit of the "horizontal" and the "vertical" curve.

to an $f$–invariant hyperbolic structure $X_1$ on $T_1$. Lift $Y_1$ to a hyperbolic structure $Y_1'$ on $T_1/\langle f^{n_2}\rangle \subset T_2/\langle f^{n_2}\rangle = Q_2$. By [22], we can find a hyperbolic structure $Y_2$ on $Q_2$ which is $f$–invariant and extends $Y_1'$. Lift $Y_2$ yields an $f$–invariant hyperbolic structure $X_2$ on $T_2$ which extends $X_1$. Repeating this procedure we obtain a sequence of hyperbolic structures $X_i$ on $T_i$ such that $X_i$ restricts to $X_{i-1}$ on $T_{i-1}$. Hence we obtain a hyperbolic structure on $D(X)$ with respect to which $f$ is a translation. By restricting the hyperbolic metric to $X$, we get the required structure. □

*Remark* 8.7. In the previous proof, in the case $e_+ = e_-$, passing to the double is necessary if $X$ has nonempty boundary. This is because the subsurfaces $T_i$ may only exhaust the interior of $X$, in which case we do not obtain a hyperbolic structure on all of $X$ (some boundary components would lie at infinity with respect to the metric on the interior of $X$).

We end this section with an example showing that the non-wandering boundary-line hypothesis is necessary in Proposition 8.3, Corollary 8.4 and Theorem 8.5.

*Example* 8.8. Let $R$ be the Roller compactification (see [23]) of $\mathbb{R}^2$ with its standard cube complex structure. Let $X$ be the surface obtained by removing from $R$ the closure of all points in $\mathbb{R}^2$ with integer coordinates. The homeomorphism $f$ of $X$ is the continuous extension of the map $(x, y) \mapsto (x + 1, y + 1)$ of $\mathbb{R}^2$. Then every curve of $X$ and every boundary line is wandering. On the other hand, there are arcs from boundary to boundary which are not wandering, such as the closure of the line $y = \frac{1}{2} \subset \mathbb{R}^2 \smallsetminus \mathbb{Z}^2$. In particular, the map induced by $f$ on $D(X)$ is not a translation.

## 9. STRUCTURE THEOREM OF WELL TEMPERED MAPS

Our goal in this section is to prove the main theorem stated in the introduction. We begin by developing several properties of the canonical decomposition associated to a well tempered map $f$. In particular, we describe how components of $S_{\mathrm{per}}$, $S_\infty$, and $S_0$ can neighbor each other. We also show that, for each component $X$ of the decomposition, the map $f$ returns to $X$. Finally, we combine these results with those of Section 8 and the work of Afton–Calegari–Chen-Lyman [3] to complete the proof of the main theorem.

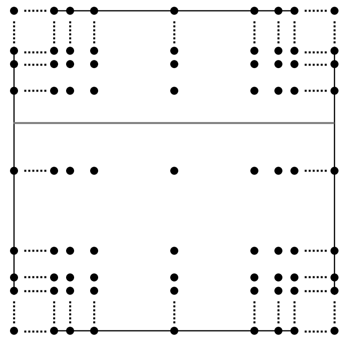

FIGURE 19. The surface $X$ from Example 8.8, with a non-wandering horizontal arc

9.1. **Properties of the canonical decomposition.** For a well tempered map, recall the almost geodesic representatives of $S_{\mathrm{per}}$, $S_\infty$, and $S_0$ defined in Section 7. An essential component of $S_{\mathrm{per}}$ or $S_\infty$ either has infinite type or has negative Euler characteristic; in particular, such a component always contains an essential curve.

Two components of the decomposition are *adjacent* if they have boundary components that are properly isotopic (equivalently, they have representatives sharing at least one boundary component).

A component $X$ is *self-adjacent* along $\alpha$ if $\alpha$ is properly homotopic to two distinct boundary components of $X$. Equivalently, $X$ is self-adjacent if there is an inessential component $Y$ which shares two boundary components with $X$.

One of the goals in this section is to show there are in fact no self-adjacent components (Lemma 9.10) and no strips (Corollary 9.11). We will also show that there is no wandering component of the decomposition, and prove that the first return map to each component is either periodic or a translation, and hence can be realized as an isometry for some hyperbolic structure with totally geodesic boundary (Lemmas 9.4, 9.5 and 9.8). Finally we determine the topological types of the components of $S_0$ (Lemma 9.12). We start by showing that certain boundary components cannot be wandering.

**Lemma 9.1.** *If two components of $S_\infty$ share a common boundary component $\alpha$, then $\alpha$ is not wandering. Similarly, if a component of $S_\infty$ is self-adjacent along $\alpha$, then $\alpha$ is not wandering.*

original proof wasn't
e right. I think it is
l now. –jt

*Proof.* First let $X$ and $Y$ be components of $S_\infty$ with a common boundary component $\alpha$. Assume for contradiction that $\alpha$ is wandering. Since both $X$ and $Y$ are essential, we can find a curve $\beta \subset X \cup Y$ that intersects $\alpha$ essentially. Because $\beta$ is not contained in $S_\infty$, it is not wandering, so $\mathcal{L}(\beta) \neq \emptyset$. Choose $\ell \in \mathcal{L}(\beta)$ and note that it must be contained in $f^n(X \cup Y)$ for some $n$; up to applying $f^{-n}$ and using $f$-invariance of limits, we may assume that $\ell \subset X \cup Y$. Let $\gamma$ be a curve intersecting $\ell$. Since $i(\gamma, \mathcal{L}(\beta)) \neq 0$, Lemma 3.7 implies $i(\mathcal{L}(\gamma), \beta) \neq 0$, hence $\gamma$ is not wandering. By Lemma 7.2, $\mathcal{L}(\gamma)$ cannot intersect $X$ or $Y$ essentially. On the other hand, $\mathcal{L}(\gamma)$ intersects $\beta$. This is only possible if $\alpha \in \mathcal{L}(\gamma)$. By $f$–invariance, the entire orbit of $\alpha$–which is infinite because $\alpha$ is wandering–lies in $\mathcal{L}(\gamma)$. This contradicts the fact that $f$ is well tempered. Therefore $\alpha$ is not wandering. □

**Lemma 9.2.** *Let $X$ be a component of $\tilde{S}_0$. If $X$ either*

- *shares at least one boundary component with $S_\infty$ but contains a non-contractible curve not homotopic to that boundary component; or*
- *shares at least two boundary components with $S_\infty$,*

*then $X$ is not wandering.*

*Proof.* In the first case, let $\gamma$ be a non-contractible curve in $X$ and let $Y$ be a component of $S_\infty$ sharing a boundary component $\alpha$ with $X$, where $\gamma$ is not homotopic to $\alpha$. Since $Y$ is essential, we can find a curve $\beta \subset X \cup Y$ crossing $\alpha$, for instance by joining a curve in $Y$ to $\gamma$ along an arc and taking the boundary of a regular neighborhood of their union. In the second case, let $Y$ and $Y'$ be components of $S_\infty$ (possibly $Y = Y'$) sharing boundary components $\alpha'$ and $\alpha''$, respectively, with $X$. We can then find a curve $\beta \subset X \cup Y \cup Y'$ intersecting $\alpha$ and $\alpha'$, by joining a curve in $Y$ to a curve in $Y'$ by an arc that crosses $\alpha$ and $\alpha'$.

We claim that if $X$ is wandering, then $\beta$ is wandering, which is a contradiction. The argument is similar to above. Indeed, suppose $\beta$ is not wandering. Then $\mathcal{L}(\beta) \neq \emptyset$, but by Lemma 7.2, $\mathcal{L}(\beta)$ cannot intersect $f^n(Y)$ or $f^n(Y')$ essentially. On the other hand, $f^n(\beta)$ is contained in $f^n(X \cup Y)$ or $f^n(X \cup Y \cup Y')$, so some line $\ell \in \mathcal{L}(\beta)$ must be homotopic into $f^n(X)$ for some $n$. Since $\mathcal{L}(\beta)$ is $f$–invariant and $X$ is wandering, infinitely many iterates of $\ell$ would lie in $\mathcal{L}(\beta)$, contradicting the finiteness of $\mathcal{L}(\beta)$. □

We now show that all boundary components of $S_\infty$ are lines.

**Lemma 9.3.** *No component $X$ of $S_\infty$ has a compact boundary component.*

*Proof.* We argue by contradiction. Suppose that $\alpha$ is a compact boundary component of $X$. Since $X$ is spanned by curves, $\alpha$ is a wandering curve. Therefore the component $Y$ adjacent to $X$ along $\alpha$ cannot belong to $S_{\mathrm{per}}$ (since boundary curves of $S_{\mathrm{per}}$ are periodic) or to $S_\infty$ (by Lemma 9.1). Hence $Y$ is a component of $S_0$.

If there is no $n > 0$ such that $f^n(Y)$ is isotopic to $Y$, then $Y$ is wandering (as $S_0$ is an $f$-invariant subsurface). By Lemma 9.2, $Y$ cannot contain a non-contractible curve different from $\alpha$. If $\alpha$ were the only boundary component of $Y$, then $Y$ is a closed disc with at most one puncture, which is impossible. Thus $Y$ has at least two boundary components, both wandering. Hence these two boundary components belong to $S_\infty$, contradicting Lemma 9.1.

Therefore there exists $n > 0$ such that $f^n(Y)$ is isotopic to $Y$. Since $\alpha$ is wandering, $Y$ contains infinitely many iterates of $\alpha$, and hence contains an essential non-peripheral curve. This contradicts Proposition 7.6, completing the proof. □

Using the previous results, we can deduce that no component of $\tilde{S}_0$ is wandering; moreover, for each such component the first return map has finite order.

**Lemma 9.4.** *For every component $X$ of $\tilde{S}_0$, $f$ returns to $X$ and there is a hyperbolic structure in which every compact boundary component has length one, such that the first return $f^k$ is isotopic to a periodic isometry of $X$.*

*Proof.* Recall that $X$ has no essential, non-peripheral curves, and by our construction of $\tilde{S}_{\mathrm{per}}$ and $S_\infty$, $X$ cannot be a closed disk or a closed disk with one puncture. If $f$ does not return to $X$, then since $\tilde{S}_0$ is a subsurface and $f$–invariant, $X$ is wandering. In particular, $X$ cannot border $\tilde{S}_{\mathrm{per}}$, so it cannot have a compact boundary by Lemma 9.3. In this case, $X$ is a disk with at most one puncture and points removed from the boundary, which yields a contradiction by Lemma 9.2. Hence $f$ returns to $X$.

Up to replacing $f$ with a power, we may assume $f(X) = X$. If $X$ has finitely many boundary components, then $f$ has finite order. Otherwise, $X$ is a disk with infinitely many points removed from the boundary and possibly a puncture or an open disk removed from its interior. Consider the action of $f$ on the noncompact boundary components of $X$. If the $f$–orbit of every component is infinite, then $f$ is semi-conjugate to an irrational rotation of the disk, which violates the fact that $f$ is well tempered. Thus, there exists a finite orbit; in that case, all periodic components of $\partial X$ have the same period.

Let $Y \subset X$ be the smallest convex subsurface containing (when present) the puncture or the compact boundary component of $X$, all the periodic boundary components of $X$, and all periodic ends of $X$. Replacing $f$ by a further power if necessary, we may assume $f$ fixes $Y$. We claim that $X = Y$. Suppose not, and let $Z$ be a component of $X \smallsetminus Y$. Topologically, $Z$ is disk with points removed from the boundary, and it shares a unique boundary line $\ell$ with $Y$. The boundary line $\ell$ is fixed by $f$, and all other boundary components of $Z$ are wandering. By the classification of orientation-preserving homeomorphism of the circle, there exist ends $e_\pm$ of $\ell$, such that for any other $\ell' \subset \partial Z$, $f^i(\ell')$ converges to $e_\pm$ as $i \to \pm\infty$.

Fix a wandering boundary component $\ell'$, and let $\ell_0$ be the line in $Z$ spanned by $e_+$ and the end of $\ell'$ such that $f^i(\ell')$, $i \geq 0$, lies on one side of $Z \smallsetminus \ell_0$. Let $Z_0$ be the subsurface corresponding to that side (see Figure 20), and let $Z_i = f^i(Z_0)$. Then $Z_i$ converges to $e_\pm$ as $i \to \pm\infty$.

Each wandering component of $\partial Z$ must be a boundary component of $S_\infty$. Thus we can find a curve $\alpha$ that intersects $\ell'$ and $X$ essentially, and $\alpha \subset Z_0 \cup S_\infty$. Such a curve must satisfy $\mathcal{L}(\alpha) \neq \emptyset$, since $\alpha$ does not belong to $S_\infty$. But this is impossible; indeed $f^i(\alpha) \subset Z_i \cup S_\infty$, while $\mathcal{L}(\alpha)$ cannot intersect $S_\infty$ essentially. Therefore $X \smallsetminus Y$ cannot have any wandering boundary components, and hence $X = Y$.

Let $k$ be the first return time to $X$. Then we obtain a hyperbolic metric on $X$ with respect to which $f^k$ is isotopic to a finite order isometry by looking at the quotient $X/\langle f^k \rangle$, choosing a hyperbolic structure with corners on the quotient and lifting it to $X$. $\square$

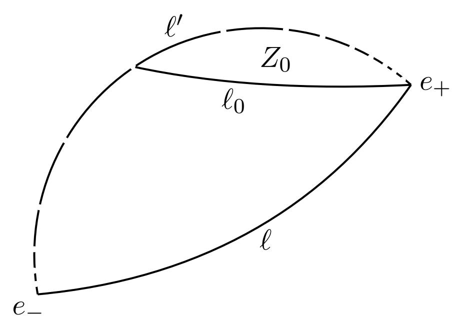


FIGURE 20. The situation in the proof of Lemma 9.4.

We can now establish the properties of the first return map for components of $S_{\text{per}}$.

**Lemma 9.5.** *Let $X$ be a component of $S_{per}$. Then $f$ returns to $X$ and there is a hyperbolic metric on $X$ with compact boundary components of length one such that the first return $f^k$ is isotopic to a periodic isometry of $X$.*

*Proof.* Since $X$ is connected and every curve in $X$ is $f$–periodic, $f$ must return to $X$. Up to replacing $f$ by a power we may assume $f(X)$ is isotopic to $X$. If $X$ has bounded topology (i.e. $X$ is finite type or the double of $X$ is of finite type), then $f$ is isotopic to a periodic map of $X$. Otherwise, choose a finite collection $\mathcal{C}$ of orbits of curves in $X$ and let $F = F(\mathcal{C})$ be the subsurface spanned by $\mathcal{C}$. After possibly enlarging $\mathcal{C}$, we can assume $F$ is connected and has negative Euler characteristic. Since $\mathcal{C}$ is $f$–invariant so is $F$; moreover, since all curves in $\mathcal{C}$ are $f$–periodic, $f|_F$ is isotopic to a periodic map of $F$ by Lemma 2.5. Let $n > 0$ be the period of $f|_F$. For any curve $\alpha$ not in $F$, we can find a larger connected $f$–invariant finite-type subsurface $F'$ containing both $F$ and $\alpha$. By the same reasoning, $f$ is also isotopic to a periodic map on $F'$, but since $F' \supset F$, $f|_{F'}$ also has period $n$ (by classical Nielsen–Thurston theory). In particular, $f^n(\alpha)$ is isotopic to $\alpha$ for all $\alpha$. By the Alexander method ([16]), this implies that $f|_X$ is isotopic to a periodic map of $X$. The fact that we can realize the first return $f^k$ to $X$ by a periodic isometry is a direct consequence of Proposition 2.7 (see [3]). $\square$

As a consequence, we can describe more precisely what changes when modifying the decomposition from $S_\infty$, $\tilde{S}_{\text{per}}$ and $\tilde{S}_0$ to $S_\infty$, $S_{\text{per}}$ and $S_0$.

**Corollary 9.6.** *Each pair of pants component of $\tilde{S}_0$ yields a pair of pants component of $S_{per}$, which contains at least one annular component of $\tilde{S}_{per}$. Furthermore, no component of $\tilde{S}_0$ is self-adjacent.*

*Proof.* Since a pair of pants component $X$ of $\tilde{S}_0$ has only compact boundary components, it must border at least one component $Y$ of $\tilde{S}_{\text{per}}$. If $Y$ is not an annulus, we can construct a curve $\alpha \subset X \cup Y$ intersecting the boundary of $X$ essentially. As some power of $f$ is periodic on both $X$ and $Y$, and by the tempered condition there is no twisting allowed along the boundary, $\alpha$ is periodic, a contradiction.

If a component $X$ of $\tilde{S}_0$ were self-adjacent along a boundary component $\alpha$, then $\alpha$ must be a curve – otherwise there is a component $Y$ which is a strip with both boundary components in $X$. Such a component $Y$ must belong to $\tilde{S}_0$, since it has no curves, but then $X \cup Y$ would be part of the same component of $\tilde{S}_0$. If $\alpha$ is a curve, then by Lemma 9.4 $\alpha$ is periodic, so the annulus $Z$ about $\alpha$ is a component of $\tilde{S}_{\text{per}}$. Since $X$ has at least two compact boundary components, it is a pair of pants, but then as before we could construct a curve $\beta$ contained in the genus-one surface $X \cup Z$ which intersects $\alpha$ essentially and is periodic, a contradiction. $\square$

Our next step is to exclude (self-)adjacency of components of $S_{\text{per}}$.

**Lemma 9.7.** *No two components of $S_{per}$ can be adjacent, neither can a component of $S_{per}$ be self-adjacent.*

*Proof.* First suppose $X$ and $Y$ are essential components of $S_{\text{per}}$ and $\alpha$ is a common boundary component. Choose a regular neighborhood $N(\alpha)$ about $\alpha$ contained in $X \cup Y$. There exists $n$ such that $f^n$ is isotopic to the identity on $X$ and $Y$. Isotope $f^n$ so that it takes $N(\alpha)$ to $N(\alpha)$ fixing $\partial N(\alpha)$ pointwise, and further isotope $f^n$ so that it's the identity on truncated subsurfaces $X \setminus N(\alpha)$ and $Y \setminus N(\alpha)$. If $N(\alpha)$ is a strip, then we can further isotope $f^n$ (rel boundary) inside of $N(\alpha)$ to the identity map. If $\alpha$ is a curve, then since $f$ is (well) tempered, it cannot twist about $\alpha$, so $f^n$ is also isotopic (rel boundary) to the identity map on $N(\alpha)$. But this implies $f^n$ is isotopic to the identity on $X \cup Y$, and in particular, we can find an essential curve $\beta$ in $X \cup Y$ crossing $\alpha$ which is $f$–periodic.

The proof that a component $X$ has no self-adjacency along a periodic component $N(\alpha)$ is similar. In this case, we can isotope $f^n$ to be identity on $X \cup N(\alpha)$, which contradicts that $X$ and $N(\alpha)$ are components of the decomposition. $\square$

With all these results at hand, we can prove the wanted properties of the first return map of components of $S_\infty$.

**Lemma 9.8.** *No component $X$ of $S_\infty$ has a wandering boundary component. In particular, $f$ returns to $X$, and there is a hyperbolic metric on $X$, such that, the first return $f^n$ is isotopic to an isometric translation on $X$.*

*Proof.* Let $\alpha$ be a wandering boundary component of $X$. Since $f$ returns to each component of $S_{\text{per}}$ and $S_0$ and the first return maps are periodic, the neighbor $Y$ of $X$ along $\alpha$ must belong to

$S_\infty$, but this contradicts Lemma 9.1. Therefore $X$ has no wandering boundary component, so $f$ must return to $X$. By Theorem 8.5, the first return $f^n$ up to isotopy preserves a hyperbolic metric on $X$ on which it acts as an isometric translation. □

Our next goal is to show that there is no self-adjacent component of the decomposition. We need a preliminary result.

**Lemma 9.9.** *A component $X$ of $S_0$ can share at most one boundary with $S_{per}$, and if $X$ shares a boundary $\alpha$ with $S_{per}$, then every non-contractible curve in $X$ is homotopic to $\alpha$.*

*Proof.* We now know $X$ is $f$–periodic (by Lemma 9.4), so the proof is similar to Lemma 9.7. The assumptions on $X$ are to rule out the existence of a periodic curve $\beta$ crossing the common boundary of $X$ and $S_{\mathrm{per}}$ essentially. □

We can now show the no self-adjacency result.

**Lemma 9.10.** *No component of the canonical decomposition is self-adjacent.*

*Proof.* Suppose $X$ is a self-adjacent component. By Corollary 9.6 and Lemma 9.7, $X$ is a component of $S_\infty$. Since $S_\infty$ has no compact boundary components by Lemma 9.3, there is a strip component $Y$ of $S_0$, with core line $\ell$, so that $X$ is the only neighbor of $Y$. Let $\ell_1$ and $\ell_2$ be the boundary components of $Y$. By Lemma 9.8, there is some $n \geq 1$ so that $f^n$ fixes $\ell_1$ (and therefore $\ell_2$, since they are homotopic). Since $Y$ is a strip, an orientation on $\ell$ induces orientations on $\ell_1$ and $\ell_2$. Now pick a curve $\alpha$ contained in $X \cup Y$ and intersecting both $\ell_1$ and $\ell_2$ once. Then $\alpha \cap X$ is an arc from $\partial X$ to $\partial X$, and by Proposition 8.3 it is wandering. This implies that $f^n$ acts as a positive translation on $\ell_1$ (with respect to the chosen orientation) if and only if it acts as a positive translation on $\ell_2$. But then we can homotope $f^n$ on the strip, leaving it invariant on its boundary, so that it is a translation, and therefore $\alpha$ is wandering, a contradiction. □

An easy consequence of the previous lemma is the fact that there are no components of the decomposition which are strips:

**Corollary 9.11.** *No component of $S_0$, $S_{per}$ or $S_\infty$ is a strip.*

*Proof.* Since by construction every component of $S_{\mathrm{per}}$ and $S_\infty$ contains an essential (possibly peripheral) curve, if there is a strip, it must be a component of $S_0$. But by construction, a strip can arise only if a component of $S_{\mathrm{per}}$ or $S_\infty$ is self-adjacent, which is impossible by Lemma 9.7 and 9.10. □

The next result is a description of the possible components of $S_0$.

**Lemma 9.12.** *The following are the topological possibilities for a component $X$ of $S_0$:*

*(1) A closed disk with at least three points removed from the boundary. In this case, at most one neighbor of $X$ is in $S_{per}$.*
*(2) A once-punctured closed disk with at least one point removed from the boundary. In this case, all neighbors of $X$ are in $S_\infty$.*
*(3) A cylinder with one compact boundary component and at least one point removed from the other boundary (a crown) In this case, $X$ has exactly one neighbor in $S_{per}$ along its compact boundary component.*

*Proof.* Note first that $X$ is not a sphere, being a subsurface of $S$. It also cannot be a disk or a once-punctured disk, by the construction of $S_\infty$ and $S_{\mathrm{per}}$. Moreover by construction $X$ has non-negative Euler characteristic, thus $\chi(X)$ is either 1 or 0.

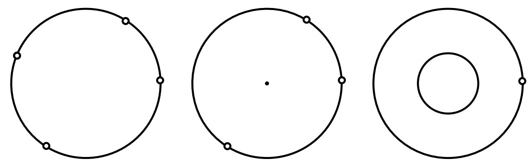

FIGURE 21. The topological possibilities for the components of $S_0$.

If $\chi(X) = 1$, then $X$ is a disk with a closed totally disconnected set with $n \in \{0, 1, \ldots \infty\}$ points removed from its boundary. If $n = 0$ or 1, $X$ would be a disk or a half-plane, which is impossible. Moreover, $n \neq 2$ by Corollary 9.11, so $n \geq 3$. By Lemma 9.9, $X$ has at most one neighbor in $S_{\mathrm{per}}$.

If $\chi(X) = 0$, then the interior of $X$ is an open annulus. We already know that $X$ cannot be the once-punctured disk, and it cannot be a closed annulus. Thus, $X$ is either a once-punctured disk or a closed annulus with at least one point removed from its boundary. If it is an annulus with points removed from both boundaries, then it would contain an essential non-peripheral curve, contradicting Proposition 7.6. If $X$ is a once-punctured disk with points removed from its boundary, then $X$ has a non-contractible curve which is non-peripheral, so it cannot border $S_{\mathrm{per}}$. If $X$ is an annulus with points removed from one of its boundary, then the core curve of $X$ is not homotopic to the non-compact boundaries of $X$, so $X$ has at most one neighbor in $S_{\mathrm{per}}$, and therefore by Lemma 9.3 it has exactly one neighbor in $S_{\mathrm{per}}$, joined along the compact boundary component. $\square$

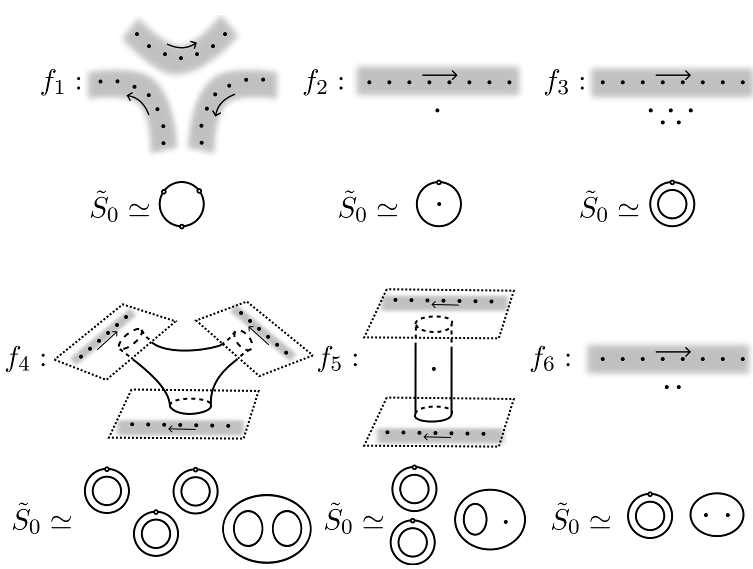


FIGURE 22. Examples of well tempered mapping classes

We end this section with some examples showing the optimality of some of our results. First, we note that all topological possibilities from Lemma 9.12 occur for components of $S_0$, and that there can be pairs of pants components of $\tilde{S}_0$. Indeed, Figure 22 exhibits examples of well tempered mapping classes with different topological types for the components of $\tilde{S}_0$ and $S_0$. In each case, the map is a puncture shift in each shaded strip, in the direction indicated by the arrow, and it's the identity outside the strips.

Finally, since a well tempered map returns to each component of $S_0$, $S_{\text{per}}$ and $S_\infty$, one might wonder if there is a uniform return time for all components. The answer is negative, as shown by the following lemma. Moreover, given a component of $S_\infty$, there is not always a uniform power of $f$ fixing all of its boundary components at once.

**Lemma 9.13.** *There is a well tempered map such that there is no uniform return time for all components of the canonical decomposition. Furthermore, the map can be chosen so that there is a component of $S_\infty$ with no uniform return time for its boundary components.*

*Proof.* Consider the bordered surface

$$P = (\mathbb{R}_{\geq 0} \times \mathbb{R}) \smallsetminus \bigcup_{n \geq 1} B_n,$$

where $B_n$ is the open disk of center $(n, 0)$ and radius $\frac{1}{4}$. Let $a_n$ be the oriented segment of the $x$-axis from $\partial B_n$ to $\partial B_{n+1}$ and $a_0$ the oriented segment of the $x$-axis from the origin to $B_1$. Define the homomorphism $\psi : \pi_1(P) \to \mathbb{Z}$ by

$$\psi(\alpha) = \sum_{n \geq 0} 2^n \hat{\imath}(\alpha, a_n),$$

where $\hat{\imath}(\cdot, \cdot)$ denotes the algebraic intersection form.

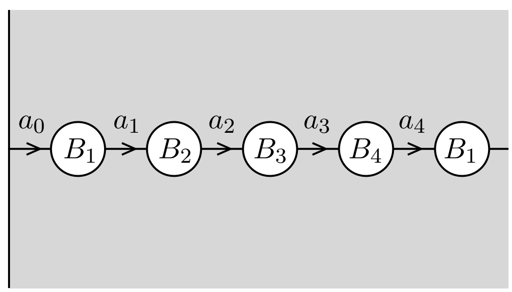


FIGURE 23. The surface $P$ with the arcs $a_n$ in the proof of Lemma 9.13.

Let $p : \Sigma \to P$ be the associated cover of the kernel of $\psi$, and let $f$ be a generator of the deck group. Then $\Sigma$ is a bordered surface, whose boundary is a union of lines, given by lifts of the $y$-axis and of the curves $\partial B_n$. Note that, for any $n$, the number of lifts of $\partial B_n$ is $2^n - 2^{n-1}$ and they are cyclically permuted by $f$. So there is no uniform power of $f$ fixing all boundary components of $\Sigma$.

To turn this into an example of a well tempered map on a borderless surface, we simply glue to each boundary component of $\Sigma$ a copy of the thrice-punctured half-plane $H$. We get a surface $S$ and a homeomorphism $\hat{f}$ of $S$ so that $\hat{f}|_\Sigma = f$ and for every copy of $H$, the first return map is the identity.

The map $\bar{f}$ is well tempered and:

- $S_\infty = \Sigma$ and $\hat{f}|_{S_\infty} = f$, so there is no bound on the first return time of the boundary components of $S_\infty$;
- $S_{\text{per}}$ is the union (over all attached copies of $H$) of components homeomorphic to a disk with three punctures, spanned by the curves in each copy of $H$. The first return time of a component equals the first return time of the boundary component of the corresponding copy of $H$, so these return times are also unbounded;

- $S_0$ is the union (over all attached copies of $H$) of components homeomorphic to an annulus with a point removed from the boundary. Again there is no uniform bound on the first return times of components of $S_0$. □

9.2. **Proof of main theorem.** Our main theorem is now an easy consequence of the results proved so far.

**Theorem 9.14.** *Let $f$ be a well tempered map of a surface $S$ of infinite type. There is a canonical decomposition of $S$ into three $f$–invariant subsurfaces $S_{per}$, $S_\infty$ and $S_0$ and a hyperbolic metric on $S$ such that for every component $X$ of $S_{per}$, $S_\infty$ and $S_0$, $f$ returns to $X$ and is isotopic to:*

- *a periodic isometry, if $X \subset S_{per} \cup S_0$,*
- *an isometric translation, if $X \subset S_\infty$.*

*Furthermore, components of $S_0$ contain no essential non-peripheral curves and at most one peripheral curve.*

*Proof.* Let $S_0$, $S_{\text{per}}$ and $S_\infty$ be given by Proposition 7.1. The hyperbolic structure on $S$ is given by gluing the hyperbolic structures provided by Lemmas 9.5 and 9.4 and Theorem 8.5. Since there are no strips (Corollary 9.11), the resulting hyperbolic structure contains no funnels or half-planes. Moreover the same results guarantee that for every component of the decomposition the first return map of $f$ exists and is isotopic to a periodic isometry or an isometric translation, as required. □

9.3. **Constructing well tempered maps.** Theorem 9.14 shows that every well tempered map is obtained by gluing together translations and periodic maps supported on disjoint subsurfaces, not accumulating anywhere. If we want to construct well tempered maps by following this procedure, we need to be careful when two supporting subsurfaces share a compact boundary component: we have to make sure that the gluing doesn't create a Dehn twist.

On the other hand, there is no need for special care when gluing along lines. More precisely:

**Lemma 9.15.** *Let $\{\Sigma_i\}_{i\in I}$ be a decomposition of a surface $S$ into essential subsurfaces without self-adjacency, not accumulating anywhere and without compact boundary components. For every $i$, let $f_i$ be either a periodic map or a translation of $\Sigma_i$, and assume that for every line $\ell \subset \partial\Sigma_i \cap \partial\Sigma_j$, $f_i(\ell) = f_j(\ell)$. Then there is a well tempered mapping class $f$ on $S$ such that the restriction of $f$ on each $\Sigma_i$ is properly homotopic to $f_i$.*

*Proof.* Define $S'$ to be the surface, homeomorphic to $S$, given by inserting strips at each line in the boundary of the decomposition. More precisely, we take the disjoint union of the $\Sigma_i$ and of a strip $S_\ell = \ell \times [0, 1]$ for every $\ell$ in the boundary, and if $\ell \subset \partial\Sigma_i \cap \partial\Sigma_j$, we glue $\ell \times \{0\}$ with the copy of $\ell$ in $\Sigma_i$ and $\ell \times \{1\}$ with the copy of $\ell$ in $\Sigma_j$. Define then $f_\ell$ on $S_\ell$ by linearly interpolating $f_i$ and $f_j$:

$$f_\ell(p, t) = (1 - t)f_i(p) + tf_j(p)$$

where we have fixed an identification of $\ell$ with $\mathbb{R}$. Then we obtain a homeomorphism $f$ of $S'$ by gluing the $f_i$ and the $f_\ell$. We claim that $f$ is well tempered.

Indeed, let $\alpha$ and $\beta$ be two curves in $S'$. By the properties of the decomposition, and up to modifying the curves by a homotopy, there are only finitely many indices $j \in I$ and lines $\ell$ in the boundary of the decomposition such that $\alpha \cap \Sigma_j$, $\beta \cap \Sigma_j$, $\alpha \cap S_\ell$ or $\beta \cap S_\ell$ are not empty. Since each $f_j$ is well tempered, the quantities

$$i(f_j^n(\alpha \cap \Sigma_j), \beta \cap \Sigma_j)$$

are uniformly bounded. Moreover,

$$i(f_\ell^n(\alpha \cap S_\ell), \beta \cap S_\ell)$$

is bounded by the number of intersection of $\alpha$ and $\beta$ with the boundary of the decomposition, hence it is also uniformly bounded. Therefore $i(f^n(\alpha), \beta)$ is uniformly bounded, so $f$ is tempered. A similar argument shows the finiteness of the limit set of $\alpha$. Hence $f$ is well tempered. □

Finally, note that if we glue periodic maps and translation and we obtain a well tempered map, the canonical decomposition might not coincide with the collection of supporting subsurfaces of the maps we are gluing. For instance, let $S = \mathbb{R}^2 \setminus \mathbb{Z}^2$, and

$$\Sigma_1 = S \cap \left\{ y \geq \frac{3}{4} \right\} \text{ and } \Sigma_2 = S \cap \left\{ y \leq \frac{1}{4} \right\}.$$

Let $f_1$ be the map $(x, y) \mapsto (x + 1, y)$ restricted to $\Sigma_1$, and $f_2$ be the map $(x, y) \to (x + 2, y)$ restricted to $\Sigma_2$ homotopic. We interpolate the maps $f_1$ and $f_2$ in the intermediate strip $\{1/4 \leq y \leq 3/4\}$. Then the induced map on $S$ is a translation; in particular, $S = S_\infty$.

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